

NCERT Exemplar Solutions For Grade 12

Mathematics

Chapter 7.- Integrals

Solved Examples

Short Answer (S.A.)

1. Integrate $\left(\frac{2a}{\sqrt{x}} - \frac{b}{x^2} + 3c\sqrt[3]{x^2}\right)$ w.r.t. x .

Ans: let $I = \int \left(\frac{2a}{\sqrt{x}} - \frac{b}{x^2} + 3c\sqrt[3]{x^2}\right) dx$

$$\Rightarrow I = 2a \int x^{-\frac{1}{2}} dx - b \int x^{-2} dx + 3c \int x^{\frac{2}{3}} dx$$

$$\Rightarrow I = 2a \times 2 \times \sqrt{x} + \frac{b}{x} + \frac{9}{5} c x^{\frac{5}{3}} + C$$

$$\Rightarrow I = 4a\sqrt{x} + \frac{b}{x} + \frac{9}{5} c x^{\frac{5}{3}} + C$$

Hence $\int \left(\frac{2a}{\sqrt{x}} - \frac{b}{x^2} + 3c\sqrt[3]{x^2}\right) dx = 4a\sqrt{x} + \frac{b}{x} + \frac{9}{5} c x^{\frac{5}{3}} + C.$

2. Evaluate $\int \frac{3ax}{b^2 + c^2 x^2} dx$

Ans: let $I = \int \frac{3ax}{b^2 + c^2 x^2} dx$

Now, put $b^2 + c^2 x^2 = t$

$$\Rightarrow 2c^2 x dx = dt$$

$$\Rightarrow x dx = \frac{dt}{2c^2}$$

$$\Rightarrow 3ax dx = \frac{3a dt}{2c^2}$$

Therefore,

$$\Rightarrow I = \frac{3a}{2c^2} \int \frac{dt}{t}$$

$$\Rightarrow I = \frac{3a}{2c^2} \log|t| + C$$

$$\Rightarrow I = \frac{3a}{2c^2} \log|b^2 + c^2 x^2| + C$$

Hence, $\int \frac{3ax}{b^2 + c^2 x^2} dx = \frac{3a}{2c^2} \log|b^2 + c^2 x^2| + C.$

3. Verify the following using the concept of integration as an antiderivative.

$$\int \frac{x^3}{x+1} dx = x - \frac{x^2}{2} + \frac{x^3}{3} - \log|x+1| + C$$

Ans: here, L.H.S $= x - \frac{x^2}{2} + \frac{x^3}{3} - \log|x+1| + C$

Now, differentiate w.r.t x , we get

$$\Rightarrow \text{L.H.S} = \frac{d}{dx} \left(x - \frac{x^2}{2} + \frac{x^3}{3} - \log|x+1| + C \right)$$

$$= 1 - \frac{2x}{2} + \frac{3x^2}{3} - \frac{1}{(x+1)} + 0$$

$$= 1 - x + x^2 - \frac{1}{(x+1)}$$

$$= \frac{(x+1) - x(x+1) + x^2(x+1) - 1}{(x+1)}$$

$$= \frac{x+1 - x^2 - x + x^3 + x^2 - 1}{(x+1)}$$

$$= \frac{x^3}{(x+1)}$$

Therefore, $\left(x - \frac{x^2}{2} + \frac{x^3}{3} - \log|x+1| + C \right) = \int \frac{x^3}{(x+1)} dx$

Hence verified.

4. Evaluate $\int \sqrt{\frac{1+x}{1-x}} dx, \quad x \neq 1.$

Ans: Let $I = \int \sqrt{\frac{1+x}{1-x}} dx$

$$\Rightarrow I = \int \sqrt{\frac{(1+x)(1+x)}{(1-x)(1+x)}} dx$$

$$\Rightarrow I = \int \sqrt{\frac{(1+x)^2}{1-x^2}} dx$$

$$\Rightarrow I = \int \frac{1+x}{\sqrt{1-x^2}} dx$$

$$\Rightarrow I = \int \frac{1}{\sqrt{1-x^2}} dx + \int \frac{x}{\sqrt{1-x^2}} dx$$

$$\Rightarrow I = \sin^{-1}x + \int \frac{x}{\sqrt{1-x^2}} dx$$

Now, put $1-x^2 = t^2$

$$\Rightarrow -2x dx = 2t dt$$

$$\Rightarrow x dx = -t dt$$

Therefore,

$$\Rightarrow I = \sin^{-1}x - \int \frac{t dt}{t}$$

$$\Rightarrow I = \sin^{-1}x - \int dt$$

$$\Rightarrow I = \sin^{-1}x - t + C$$

$$\Rightarrow I = \sin^{-1}x - \sqrt{1-x^2} + C$$

Hence, $\int \frac{\sqrt{1+x}}{\sqrt{1-x}} dx = \sin^{-1}x - \sqrt{1-x^2} + C$

5. Evaluate $\int \frac{dx}{\sqrt{(x-\alpha)(\beta-x)}}, \beta > \alpha$

Ans: Let $I = \int \frac{dx}{\sqrt{(x-\alpha)(\beta-x)}}$

Now, put $x - \alpha = t^2$

$$\Rightarrow dx = 2t dt$$

And $\beta - x = \beta - (t^2 + \alpha) = \beta - \alpha - t^2$

Therefore,

$$\Rightarrow I = \int \frac{2t dt}{\sqrt{t^2(\beta - \alpha - t^2)}}$$

$$\Rightarrow I = 2 \int \frac{dt}{\sqrt{(\beta - \alpha - t^2)}}$$

$$\Rightarrow I = 2 \int \frac{dt}{\sqrt{\left[(\sqrt{\beta - \alpha})^2 - t^2\right]}}$$

$$\Rightarrow I = 2 \sin^{-1} \frac{t}{\sqrt{\beta - \alpha}} + C$$

$$\Rightarrow I = 2 \sin^{-1} \frac{\sqrt{x - \alpha}}{\sqrt{\beta - \alpha}} + C$$

Hence, $\int \frac{dx}{\sqrt{(x - \alpha)(\beta - x)}} = 2 \sin^{-1} \frac{\sqrt{x - \alpha}}{\sqrt{\beta - \alpha}} + C.$

6. Evaluate $\int \tan^8 x \sec^4 x \, dx$

Ans: let $I = \int \tan^8 x \sec^4 x \, dx$

$$\Rightarrow I = \int \tan^8 x (\sec^2 x) \sec^2 x \, dx$$

$$\Rightarrow I = \int \tan^8 x (1 + \tan^2 x) \sec^2 x \, dx$$

Now, put $\tan x = t$

$$\Rightarrow \sec^2 x \, dx = dt$$

Therefore,

$$\Rightarrow I = \int t^8 (1 + t^2) dt$$

$$\Rightarrow I = \int (t^8 + t^{10}) dt$$

$$\Rightarrow I = \frac{t^9}{9} + \frac{t^{11}}{11} + C$$

$$\Rightarrow I = \frac{\tan^9 x}{9} + \frac{\tan^{11} x}{11} + C$$

Hence, $\int \tan^8 x \sec^4 x dx = \frac{\tan^9 x}{9} + \frac{\tan^{11} x}{11} + C$

7. Find $\int \frac{x^3}{x^4 + 3x^2 + 2} dx$

Ans: Let $I = \int \frac{x^3}{x^4 + 3x^2 + 2} dx$

$\Rightarrow I = \int \frac{x^2 \cdot x dx}{x^4 + 3x^2 + 2}$

Now, put $x^2 = t$

$\Rightarrow 2x dx = dt$

$\Rightarrow x dx = \frac{dt}{2}$

Therefore,

$\Rightarrow I = \frac{1}{2} \int \frac{t dt}{t^2 + 3t + 2}$

$\Rightarrow I = \frac{1}{2} \int \frac{t dt}{(t+1)(t+2)}$

Consider $\frac{t}{(t+1)(t+2)} = \frac{A}{t+1} + \frac{B}{t+2}$

$\Rightarrow \frac{t}{(t+1)(t+2)} = \frac{At + 2A + Bt + B}{(t+1)(t+2)}$

$\Rightarrow \frac{t}{(t+1)(t+2)} = \frac{(A+B)t + 2A+B}{(t+1)(t+2)}$

$\Rightarrow t = (A+B)t + 2A+B$

On comparing both sides, we get

$$\Rightarrow A+B=1 \text{ and } 2A+B=0$$

On solving, we get

$$\Rightarrow A=-1, B=2$$

Therefore,

$$\Rightarrow I = \frac{1}{2} \int \frac{t dt}{(t+1)(t+2)}$$

$$\Rightarrow I = \frac{1}{2} \int \left[\frac{-1}{t+1} + \frac{2}{t+2} \right] dt$$

$$\Rightarrow I = \frac{1}{2} (-\log|t+1| + 2\log|t+2|) + C$$

$$\Rightarrow I = -\frac{1}{2} \log|t+1| + \log|t+2| + C$$

$$\Rightarrow I = -\log|\sqrt{t+1}| + \log|t+2| + C$$

$$\Rightarrow I = \log \left| \frac{t+2}{\sqrt{t+1}} \right| + C$$

$$\Rightarrow I = \log \left| \frac{x^2+2}{\sqrt{x^2+1}} \right| + C$$

$$\text{Hence, } I = \int \frac{x^3}{x^4+3x^2+2} dx = \log \left| \frac{x^2+2}{\sqrt{x^2+1}} \right| + C$$

8. Find $\int \frac{dx}{2 \sin^2 x + 5 \cos^2 x}$

Ans: Let $I = \int \frac{dx}{2 \sin^2 x + 5 \cos^2 x}$

Dividing numerator and denominator by $\cos^2 x$, we get

$$\Rightarrow I = \int \frac{\sec^2 x dx}{2 \tan^2 x + 5}$$

Put $\tan x = t$

$$\Rightarrow \sec^2 x dx = dt$$

Therefore,

$$\Rightarrow I = \int \frac{dt}{2t^2 + 5}$$

$$\Rightarrow I = \frac{1}{2} \int \frac{dt}{t^2 + \frac{5}{2}}$$

$$\Rightarrow I = \frac{1}{2} \int \frac{dt}{t^2 + \left(\sqrt{\frac{5}{2}}\right)^2}$$

$$\Rightarrow I = \frac{1}{2} \times \sqrt{\frac{2}{5}} \tan^{-1} \frac{\sqrt{2}t}{\sqrt{5}} + C$$

$$\Rightarrow I = \frac{1}{\sqrt{10}} \tan^{-1} \left(\frac{\sqrt{2} \tan x}{\sqrt{5}} \right) + C$$

Hence, $\int \frac{dx}{2 \sin^2 x + 5 \cos^2 x} = \frac{1}{\sqrt{10}} \tan^{-1} \left(\frac{\sqrt{2} \tan x}{\sqrt{5}} \right) + C.$

9. Evaluate $\int_{-1}^2 (7x - 5) dx$ as a limit of sums.

Ans: here, we have $a = -1$, $b = 2$ and $h = \frac{2+1}{n} \Rightarrow nh = 3$

And $f(x) = 7x - 5$

Now, we have

$$\Rightarrow \int_{-1}^2 (7x - 5) dx = \lim_{h \rightarrow 0} h [f(-1) + f(-1+h) + f(-1+2h) \dots + f(-1+(n-1)h)]$$

Also we have,

$$\Rightarrow f(-1) = -7 - 5 = -12$$

$$\Rightarrow f(-1+h) = -7 + 7h - 5 = -12 + 7h$$

$$\Rightarrow f(-1+(n-1)h) = 7(n-1)h - 12 =$$

Therefore,

$$\Rightarrow \int_{-1}^2 (7x - 5) dx = \lim_{h \rightarrow 0} h [-12 + (-7h - 12) + (-14h - 12) \dots + (7(n-1)h - 12)]$$

$$\Rightarrow \int_{-1}^2 (7x - 5) dx = \lim_{h \rightarrow 0} h [7h[1 + 2 + \dots + (n-10)] - 12n]$$

$$\Rightarrow \int_{-1}^2 (7x - 5) dx = \lim_{h \rightarrow 0} h \left[7h \times \frac{n(n-1)}{2} - 12n \right]$$

$$\Rightarrow \int_{-1}^2 (7x - 5) dx = \lim_{h \rightarrow 0} h \left[\frac{7}{2} (nh)(nh - h) - 12nh \right]$$

$$\Rightarrow \int_{-1}^2 (7x - 5) dx = \frac{7}{2} \times 3 \times 3 - 12 \times 3$$

$$\Rightarrow \int_{-1}^2 (7x - 5) dx = \frac{63}{2} - 36$$

$$\Rightarrow \int_{-1}^2 (7x - 5) dx = \frac{63 - 72}{2}$$

$$\Rightarrow \int_{-1}^2 (7x - 5) dx = \frac{-9}{2}$$

Hence, $\int_{-1}^2 (7x - 5) dx = \frac{-9}{2}.$

10. Evaluate $\int_0^{\frac{\pi}{2}} \frac{\tan^7 x}{\cot^7 x + \tan^7 x} dx$

Ans: Let $I = \int_0^{\frac{\pi}{2}} \frac{\tan^7 x}{\cot^7 x + \tan^7 x} dx \dots\dots\dots (i)$

$\Rightarrow I = \int_0^{\frac{\pi}{2}} \frac{\tan^7 \left(\frac{\pi}{2} - x \right)}{\cot^7 \left(\frac{\pi}{2} - x \right) + \tan^7 \left(\frac{\pi}{2} - x \right)} dx$

$\Rightarrow I = \int_0^{\frac{\pi}{2}} \frac{\cot^7 x}{\tan^7 x + \cot^7 x} dx \dots\dots\dots (ii)$

Now, adding eq (i) and (ii), we get

$\Rightarrow 2I = \int_0^{\frac{\pi}{2}} \frac{\tan^7 x + \cot^7 x}{\tan^7 x + \cot^7 x} dx$

$\Rightarrow 2I = \int_0^{\frac{\pi}{2}} dx$

$\Rightarrow 2I = [x]_0^{\frac{\pi}{2}}$

$\Rightarrow 2I = \frac{\pi}{2}$

$\Rightarrow I = \frac{\pi}{4}$

Hence, $\int_0^{\frac{\pi}{2}} \frac{\tan^7 x}{\cot^7 x + \tan^7 x} dx = \frac{\pi}{4}.$

11. Find $\int_2^8 \frac{\sqrt{10-x}}{\sqrt{x} + \sqrt{10-x}} dx$

Ans: Let $I = \int_2^8 \frac{\sqrt{10-x}}{\sqrt{x} + \sqrt{10-x}} dx$ (i)

$$\Rightarrow I = \int_2^8 \frac{\sqrt{8+2-(10-x)}}{\sqrt{8+2-x} + \sqrt{8+2-(10-x)}} dx$$

$$\Rightarrow I = \int_2^8 \frac{\sqrt{10-10+x}}{\sqrt{10-x} + \sqrt{10-10+x}} dx$$

$$\Rightarrow I = \int_2^8 \frac{\sqrt{x}}{\sqrt{10-x} + \sqrt{x}} dx$$
 (ii)

Now, adding eq (i) and (ii), we get

$$\Rightarrow 2I = \int_2^8 \frac{\sqrt{10-x} + \sqrt{x}}{\sqrt{10-x} + \sqrt{x}} dx$$

$$\Rightarrow 2I = \int_2^8 dx$$

$$\Rightarrow 2I = [x]_2^8$$

$$\Rightarrow 2I = 8 - 2$$

$$\Rightarrow 2I = 6$$

$$\Rightarrow I = 3$$

Hence, $\int_2^8 \frac{\sqrt{10-x}}{\sqrt{x} + \sqrt{10-x}} dx = 3.$

12. Find $\int_0^{\frac{\pi}{4}} \sqrt{1 + \sin 2x} \, dx$

Ans: Let $I = \int_0^{\frac{\pi}{4}} \sqrt{1 + \sin 2x} \, dx$

$$\Rightarrow I = \int_0^{\frac{\pi}{4}} \sqrt{\sin^2 x + \cos^2 x + 2 \sin x \cdot \cos x} \, dx$$

$$\Rightarrow I = \int_0^{\frac{\pi}{4}} \sqrt{(\sin x + \cos x)^2} \, dx$$

$$\Rightarrow I = \int_0^{\frac{\pi}{4}} (\sin x + \cos x) \, dx$$

$$\Rightarrow I = [-\cos x + \sin x]_0^{\frac{\pi}{4}}$$

$$\Rightarrow I = \left[-\cos \frac{\pi}{4} + \sin \frac{\pi}{4} - (-\cos 0 + \sin 0) \right]$$

$$\Rightarrow I = \left[-\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} - (-1 + 0) \right]$$

$$\Rightarrow I = 1$$

Hence, $\int_0^{\frac{\pi}{4}} \sqrt{1 + \sin 2x} \, dx = 1.$

13. Find $\int x^2 \tan^{-1} x \, dx$

Ans: Let $I = \int x^2 \tan^{-1} x \, dx$

$$\Rightarrow I = \tan^{-1} x \int x^2 \, dx - \int \left(\frac{d}{dx} (\tan^{-1} x) \int x^2 \, dx \right) dx$$

$$\Rightarrow I = \frac{x^3}{3} \tan^{-1} x - \int \frac{1}{1+x^2} \cdot \frac{x^3}{3} dx$$

$$\Rightarrow I = \frac{x^3}{3} \tan^{-1} x - \frac{1}{3} \int \frac{x^3}{1+x^2} dx$$

$$\Rightarrow I = \frac{x^3}{3} \tan^{-1} x - \frac{1}{3} \int \left(x - \frac{x}{1+x^2} \right) dx$$

$$\Rightarrow I = \frac{x^3}{3} \tan^{-1} x - \frac{1}{3} \int x dx + \frac{1}{3} \int \frac{x}{1+x^2} dx$$

$$\Rightarrow I = \frac{x^3}{3} \tan^{-1} x - \frac{x^2}{6} + \frac{1}{3} \int \frac{x}{1+x^2} dx$$

Now, put $1+x^2 = t$

$$\Rightarrow 2x dx = dt$$

$$\Rightarrow x dx = \frac{dt}{2}$$

$$\Rightarrow I = \frac{x^3}{3} \tan^{-1} x - \frac{x^2}{6} + \frac{1}{6} \int \frac{dt}{t}$$

$$\Rightarrow I = \frac{x^3}{3} \tan^{-1} x - \frac{x^2}{6} + \frac{1}{6} \log|t| + C$$

$$\Rightarrow I = \frac{x^3}{3} \tan^{-1} x - \frac{x^2}{6} + \frac{1}{6} \log|1+x^2| + C$$

14. Find $\int \sqrt{10-4x+4x^2} dx$

Ans: Let $I = \int \sqrt{10-4x+4x^2} dx$

$$\Rightarrow I = \int \sqrt{4x^2 - 4x + 10} dx$$

$$\Rightarrow I = \int \sqrt{4\left(x^2 - x + \frac{10}{4}\right)} dx$$

$$\Rightarrow I = 2 \int \sqrt{\left[x^2 - x + \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2 + \frac{10}{4}\right]} dx$$

$$\Rightarrow I = 2 \int \sqrt{\left[\left(x - \frac{1}{2}\right)^2 - \frac{1}{4} + \frac{10}{4}\right]} dx$$

$$\Rightarrow I = 2 \int \sqrt{\left[\left(x - \frac{1}{2}\right)^2 + \frac{9}{4}\right]} dx$$

$$\Rightarrow I = 2 \int \sqrt{\left[\left(x - \frac{1}{2}\right)^2 + \left(\frac{3}{2}\right)^2\right]} dx$$

$$\Rightarrow I = \left[\frac{\left(x - \frac{1}{2}\right)}{2} \sqrt{4x^2 - 4x + 10} + \frac{\left(\frac{3}{2}\right)^2}{2} \log \left| \left(x - \frac{1}{2}\right) + \sqrt{4x^2 - 4x + 10} \right| \right]$$

$$\Rightarrow I = \left(\frac{2x-1}{4} \right) \sqrt{4x^2 - 4x + 10} + \frac{9}{4} \log \left| \left(x - \frac{1}{2}\right) + \sqrt{4x^2 - 4x + 10} \right|$$

Long Answer (L.A.)

15. Evaluate $\int \frac{x^2 dx}{x^4 + x^2 - 2}$

Ans: Let $I = \int \frac{x^2 dx}{x^4 + x^2 - 2}$

Now, put $x^2 = t$

Therefore,

$$\Rightarrow \frac{x^2}{x^4 + x^2 - 2} = \frac{t}{t^2 + t - 2} = \frac{t}{(t+2)(t-1)}$$

$$\Rightarrow \frac{t}{(t+2)(t-1)} = \frac{A}{t+2} + \frac{B}{t-1}$$

$$\Rightarrow \frac{t}{(t+2)(t-1)} = \frac{At - A + Bt + 2B}{(t+2)(t-1)}$$

$$\Rightarrow t = (A+B)t + 2B - A$$

On comparing both side, we get

$$\Rightarrow A+B=1 \text{ and } 2B-A=0$$

On solving these equations, we get

$$\Rightarrow B = \frac{1}{3} \text{ and } A = \frac{2}{3}$$

$$\text{So, } \frac{x^2}{x^4 + x^2 - 2} = \frac{2}{3} \frac{1}{x^2 + 2} + \frac{1}{3} \frac{1}{x^2 - 1}$$

Therefore,

$$\Rightarrow I = \int \left(\frac{2}{3} \frac{1}{x^2 + 2} + \frac{1}{3} \frac{1}{x^2 - 1} \right) dx$$

$$\Rightarrow I = \frac{2}{3} \int \frac{1}{x^2 + (\sqrt{2})^2} dx + \frac{1}{3} \int \frac{1}{x^2 - 1} dx$$

$$\Rightarrow I = \frac{2}{3} \times \frac{1}{\sqrt{2}} \tan^{-1} \frac{x}{\sqrt{2}} + \frac{1}{3} \times \frac{1}{2} \log \left| \frac{x-1}{x+1} \right| + C$$

$$\Rightarrow I = \frac{\sqrt{2}}{3} \tan^{-1} \frac{x}{\sqrt{2}} + \frac{1}{6} \log \left| \frac{x-1}{x+1} \right| + C$$

$$\text{Hence, } \int \frac{x^2 dx}{x^4 + x^2 - 2} = \frac{\sqrt{2}}{3} \tan^{-1} \frac{x}{\sqrt{2}} + \frac{1}{6} \log \left| \frac{x-1}{x+1} \right| + C.$$

16. Evaluate $\int \frac{x^3 + x}{x^4 - 9} dx$

Ans: Let $I = \int \frac{x^3 + x}{x^4 - 9} dx$

$$\Rightarrow I = \int \frac{x^3}{x^4 - 9} dx + \int \frac{x}{x^4 - 9} dx$$

$$\Rightarrow I = I_1 + I_2 \dots\dots\dots (i)$$

where $I_1 = \int \frac{x^3}{x^4 - 9} dx$ and $I_2 = \int \frac{x}{x^4 - 9} dx$

Now, $I_1 = \int \frac{x^3}{x^4 - 9} dx$

Put $x^4 - 9 = t$

$$\Rightarrow 4x^3 dx = dt$$

$$\Rightarrow x^3 dx = \frac{dt}{4}$$

Therefore,

$$\Rightarrow I_1 = \frac{1}{4} \int \frac{dt}{t}$$

$$\Rightarrow I_1 = \frac{1}{4} \log|t| + C_1$$

$$\Rightarrow I_1 = \frac{1}{4} \log|x^4 - 9| + C_1 \dots\dots\dots (ii)$$

Now, $I_2 = \int \frac{x}{x^4 - 9} dx$

Put $x^2 = t$

$$\Rightarrow 2x dx = dt$$

$$\Rightarrow x dx = \frac{dt}{2}$$

Therefore,

$$\Rightarrow I_2 = \frac{1}{2} \int \frac{dt}{t^2 - 3^2}$$

$$\Rightarrow I_2 = \frac{1}{2} \times \frac{1}{2 \times 3} \log \left| \frac{t-3}{t+3} \right| + C_2$$

$$\Rightarrow I_2 = \frac{1}{12} \log \left| \frac{x^2 - 3}{x^2 + 3} \right| + C_2$$

Now, put the value of I_1 and I_2 in eq (i), we get

$$\Rightarrow I = \frac{1}{4} \log |x^4 - 9| + C_1 + \frac{1}{12} \log \left| \frac{x^2 - 3}{x^2 + 3} \right| + C_2$$

$$\Rightarrow I = \frac{1}{4} \log |x^4 - 9| + \frac{1}{12} \log \left| \frac{x^2 - 3}{x^2 + 3} \right| + C$$

17. Show that $\int_0^{\frac{\pi}{2}} \frac{\sin^2 x dx}{\sin x + \cos x} = \frac{1}{\sqrt{2}} \log(\sqrt{2} + 1)$

Ans: let $I = \int_0^{\frac{\pi}{2}} \frac{\sin^2 x dx}{\sin x + \cos x}$ (i)

$$\Rightarrow I = \int_0^{\frac{\pi}{2}} \frac{\sin^2 \left(\frac{\pi}{2} - x \right) dx}{\sin \left(\frac{\pi}{2} - x \right) + \cos \left(\frac{\pi}{2} - x \right)}$$

$$\Rightarrow I = \int_0^{\frac{\pi}{2}} \frac{\cos^2 x \, dx}{\cos x + \sin x} \dots\dots\dots (ii)$$

Adding eq (i) and (ii), we get

$$\Rightarrow 2I = \int_0^{\frac{\pi}{2}} \frac{(\sin^2 x + \cos^2 x) \, dx}{\cos x + \sin x}$$

$$\Rightarrow 2I = \int_0^{\frac{\pi}{2}} \frac{dx}{\cos x + \sin x}$$

$$\Rightarrow 2I = \frac{1}{\sqrt{2}} \int_0^{\frac{\pi}{2}} \frac{dx}{\cos x \cdot \frac{1}{\sqrt{2}} + \sin x \cdot \frac{1}{\sqrt{2}}}$$

$$\Rightarrow 2I = \frac{1}{\sqrt{2}} \int_0^{\frac{\pi}{2}} \frac{dx}{\cos x \cos \frac{\pi}{4} + \sin x \sin \frac{\pi}{4}}$$

$$\Rightarrow 2I = \frac{1}{\sqrt{2}} \int_0^{\frac{\pi}{2}} \frac{dx}{\cos \left(x - \frac{\pi}{4} \right)}$$

$$\Rightarrow 2I = \frac{1}{\sqrt{2}} \int_0^{\frac{\pi}{2}} \sec \left(x - \frac{\pi}{4} \right) dx$$

$$\Rightarrow 2I = \frac{1}{\sqrt{2}} \left[\log \left\{ \sec \left(x - \frac{\pi}{4} \right) + \tan \left(x - \frac{\pi}{4} \right) \right\} \right]_0^{\frac{\pi}{2}}$$

$$\Rightarrow 2I = \frac{1}{\sqrt{2}} \left[\log \left\{ \sec \left(\frac{\pi}{2} - \frac{\pi}{4} \right) + \tan \left(\frac{\pi}{2} - \frac{\pi}{4} \right) \right\} - \log \left\{ \sec \left(-\frac{\pi}{4} \right) + \tan \left(-\frac{\pi}{4} \right) \right\} \right]$$

$$\Rightarrow 2I = \frac{1}{\sqrt{2}} \left[\log \left\{ \sec \left(\frac{\pi}{4} \right) + \tan \left(\frac{\pi}{4} \right) \right\} - \log \left\{ \sec \left(\frac{\pi}{4} \right) - \tan \left(\frac{\pi}{4} \right) \right\} \right]$$

$$\Rightarrow 2I = \frac{1}{\sqrt{2}} [\log(\sqrt{2} + 1) - \log(\sqrt{2} - 1)]$$

$$\Rightarrow 2I = \frac{1}{\sqrt{2}} \times \log \left| \frac{\sqrt{2} + 1}{\sqrt{2} - 1} \right|$$

$$\Rightarrow 2I = \frac{1}{\sqrt{2}} \times \log \left| \frac{(\sqrt{2} + 1)^2}{2 - 1} \right|$$

$$\Rightarrow 2I = \frac{1}{\sqrt{2}} \times 2 \log(\sqrt{2} + 1)$$

$$\Rightarrow I = \frac{1}{\sqrt{2}} \log(\sqrt{2} + 1)$$

Hence proved

18. Find $\int_0^1 x (\tan^{-1} x)^2 dx$

Ans: Let $I = \int_0^1 x (\tan^{-1} x)^2 dx$

Applying integrating by parts, we get

$$\Rightarrow I = \left[\frac{x^2}{2} (\tan^{-1} x)^2 \right]_0^1 - \int_0^1 \frac{2 \tan^{-1} x}{1 + x^2} \cdot \frac{x^2}{2} dx$$

$$\Rightarrow I = \left[\frac{1}{2} (\tan^{-1} 1)^2 - 0 \right] - \int_0^1 \frac{\tan^{-1} x}{1 + x^2} x^2 dx$$

$$\Rightarrow I = \left[\frac{1}{2} \left(\frac{\pi}{4} \right)^2 \right] - \int_0^1 \frac{(x^2 + 1 - 1) \tan^{-1} x}{1 + x^2} dx$$

$$\Rightarrow I = \frac{\pi^2}{32} - \int_0^1 \frac{(x^2 + 1 - 1) \tan^{-1} x}{1 + x^2} dx$$

$$\Rightarrow I = \frac{\pi^2}{32} - \int_0^1 \frac{(x^2 + 1)\tan^{-1}x}{1+x^2} dx + \int_0^1 \frac{\tan^{-1}x}{1+x^2} dx$$

$$\Rightarrow I = \frac{\pi^2}{32} - \int_0^1 \tan^{-1}x dx + \int_0^1 \frac{\tan^{-1}x}{1+x^2} dx$$

$$\Rightarrow I = \frac{\pi^2}{32} - I_1 + I_2 \quad \dots\dots\dots (i)$$

Where, $I_1 = \int_0^1 \tan^{-1}x dx$ and $I_2 = \int_0^1 \frac{\tan^{-1}x}{1+x^2} dx$

Now, $I_1 = \int_0^1 \tan^{-1}x dx$

$$\Rightarrow I_1 = [x \tan^{-1}x]_0^1 - \int_0^1 \frac{x}{1+x^2} dx$$

$$\Rightarrow I_1 = [1 \cdot \tan^{-1}1 - 0] - \int_0^1 \frac{x}{1+x^2} dx$$

$$\Rightarrow I_1 = \frac{\pi}{4} - \int_0^1 \frac{x}{1+x^2} dx$$

Now, put $1+x^2 = t$

$$\Rightarrow 2x dx = dt$$

$$\Rightarrow x dx = \frac{dt}{2}$$

And at $x=0 \rightarrow t=1$

$$\Rightarrow \text{at } x=1 \rightarrow t=2$$

Therefore,

$$\Rightarrow I_1 = \frac{\pi}{4} - \frac{1}{2} \int_1^2 \frac{dt}{t}$$

$$\Rightarrow I_1 = \frac{\pi}{4} - \frac{1}{2} [\log t]_1^2$$

$$\Rightarrow I_1 = \frac{\pi}{4} - \frac{1}{2} [\log 2 - \log 1]$$

$$\Rightarrow I_1 = \frac{\pi}{4} - \frac{1}{2} \log 2 \dots\dots\dots (ii)$$

Now, $I_2 = \int_0^1 \frac{\tan^{-1} x}{1+x^2} dx$

Put $\tan^{-1} x = t$

$$\Rightarrow \frac{1}{1+x^2} dx = dt$$

And at $x=0 \rightarrow t=0$

$$\Rightarrow \text{at } x=1 \rightarrow t = \frac{\pi}{4}$$

Therefore,

$$\Rightarrow I_2 = \int_0^{\frac{\pi}{4}} t dt$$

$$\Rightarrow I_2 = \left[\frac{t^2}{2} \right]_0^{\frac{\pi}{4}}$$

$$\Rightarrow I_2 = \frac{1}{2} \left[t^2 \right]_0^{\frac{\pi}{4}}$$

$$\Rightarrow I_2 = \frac{1}{2} \left[\frac{\pi^2}{16} - 0 \right]$$

$$\Rightarrow I_2 = \frac{\pi^2}{32}$$

Now, put the value of I_2 and I_2 in eq(i), we get

$$\Rightarrow I = \frac{\pi^2}{32} - \frac{\pi}{4} + \frac{1}{2} \log 2 + \frac{\pi^2}{32}$$

$$\Rightarrow I = \frac{\pi^2}{16} - \frac{\pi}{4} + \frac{1}{2} \log 2$$

Hence, $\int_0^1 x (\tan^{-1} x)^2 dx = \frac{\pi^2}{16} - \frac{\pi}{4} + \frac{1}{2} \log 2.$

19. Evaluate $\int_{-1}^2 f(x) dx$, where $f(x) = |x+1| + |x| + |x-1|$.

Ans: here, we can redefine $f(x)$ as $f(x) = \begin{cases} 2-x, & \text{if } -1 < x \leq 0 \\ x+2, & \text{if } 0 < x \leq 1 \\ 3x, & \text{if } 1 < x \leq 2 \end{cases}$

Therefore, $\int_{-1}^2 f(x) dx = \int_{-1}^0 (2-x) dx + \int_0^1 (x+2) dx + \int_1^2 3x dx$

$$\Rightarrow \int_{-1}^2 f(x) dx = \left[2x - \frac{x^2}{2} \right]_{-1}^0 + \left[\frac{x^2}{2} + 2x \right]_0^1 + 3 \left[\frac{x^2}{2} \right]_1^2$$

$$\Rightarrow \int_{-1}^2 f(x) dx = \left[0 - \left(-2 - \frac{1}{2} \right) \right] + \left[\left(\frac{1}{2} + 2 \right) - 0 \right] + 3 \left[2 - \frac{1}{2} \right]$$

$$\Rightarrow \int_{-1}^2 f(x) dx = \frac{5}{2} + \frac{5}{2} + \frac{9}{2}$$

$$\Rightarrow \int_{-1}^2 f(x) dx = \frac{19}{2}$$

Objective Type Questions

Choose the correct answer from the given four option in each of the Examples from 20 to 30.

20. $\int e^x (\cos x - \sin x) dx$ is equal to

- (A) $e^x \cos x + C$
- (B) $e^x \sin x + C$
- (C) $-e^x \cos x + C$
- (D) $-e^x \sin x + C$

Ans: here, as we know that $\int e^x [f(x) + f'(x)] dx = e^x f(x) + C$

Now, if $f(x) = \cos x$, then $f'(x) = -\sin x$

Therefore, $\int e^x (\cos x - \sin x) dx = e^x \cos x + C$

Hence, option (A) is the correct answer.

21. $\int \frac{dx}{\sin^2 x \cos^2 x}$ is equal to

- (A) $\tan x + \cot x + C$
- (B) $(\tan x + \cot x)^2 + C$
- (C) $\tan x - \cot x + C$
- (D) $(\tan x - \cot x)^2 + C$

Ans: Let $I = \int \frac{dx}{\sin^2 x \cos^2 x}$

$$\Rightarrow I = \int \frac{(\sin^2 x + \cos^2 x) dx}{\sin^2 x \cos^2 x}$$

$$\Rightarrow I = \int \frac{\sin^2 x dx}{\sin^2 x \cos^2 x} + \int \frac{\cos^2 x dx}{\sin^2 x \cos^2 x}$$

$$\Rightarrow I = \int \frac{dx}{\cos^2 x} + \int \frac{dx}{\sin^2 x}$$

$$\Rightarrow I = \int \sec^2 x dx + \int \operatorname{cosec}^2 x dx$$

$$\Rightarrow I = \tan x - \cot x + C$$

Hence, option (C) is the correct answer.

22. If $\int \frac{3e^x - 5e^{-x}}{4e^x + 5e^{-x}} dx = ax + b \log|4e^x + 5e^{-x}| + C$, then

(A) $a = -\frac{1}{8}, b = \frac{7}{8}$

(B) $a = \frac{1}{8}, b = \frac{7}{8}$

(C) $a = -\frac{1}{8}, b = -\frac{7}{8}$

(D) $a = \frac{1}{8}, b = -\frac{7}{8}$

Ans: here, we have $\int \frac{3e^x - 5e^{-x}}{4e^x + 5e^{-x}} dx = ax + b \log|4e^x + 5e^{-x}| + C$

Now, differentiating both sides, we have

$$\Rightarrow \frac{d}{dx} \left(\int \frac{3e^x - 5e^{-x}}{4e^x + 5e^{-x}} dx \right) = \frac{d}{dx} (ax + b \log|4e^x + 5e^{-x}| + C)$$

$$\Rightarrow \frac{3e^x - 5e^{-x}}{4e^x + 5e^{-x}} = a + b \times \frac{1}{4e^x + 5e^{-x}} \times (4e^x - 5e^{-x})$$

$$\Rightarrow \frac{3e^x - 5e^{-x}}{4e^x + 5e^{-x}} = a + b \left(\frac{4e^x - 5e^{-x}}{4e^x + 5e^{-x}} \right)$$

$$\Rightarrow (3e^x - 5e^{-x}) = a(4e^x + 5e^{-x}) + b(4e^x - 5e^{-x})$$

$$\Rightarrow (3e^x - 5e^{-x}) = 4ae^x + 5ae^{-x} + 4be^x - 5be^{-x}$$

$$\Rightarrow 3e^x - 5e^{-x} = 4e^x(a + b) - 5e^{-x}(-a + b)$$

Now, comparing both sides, we get

$$\Rightarrow 4(a + b) = 3 \text{ and } (-a + b) = 1$$

$$\Rightarrow (a + b) = \frac{3}{4} \dots\dots (i) \text{ and } (-a + b) = 1 \dots\dots\dots(ii)$$

Adding eq (i) and (ii), we get

$$\Rightarrow 2b = \frac{7}{4}$$

$$\Rightarrow b = \frac{7}{8}.$$

Now, put the value of b in eq (ii),

$$\Rightarrow -a + \frac{7}{8} = 1$$

$$\Rightarrow -a = 1 - \frac{7}{8}$$

$$\Rightarrow a = -\frac{1}{8}$$

Here, $a = -\frac{1}{8}$ and $b = \frac{7}{8}$

Hence, option (A) is the correct answer.

23: $\int_{a+c}^{b+c} f(x) dx$ is equal to

(A) $\int_a^b f(x-c) dx$

(B) $\int_a^b f(x+c) dx$

(C) $\int_a^b f(x) dx$

(D) $\int_{a-c}^{b-c} f(x) dx$

Ans: Let $I = \int_{a+c}^{b+c} f(x) dx$

Now, put $x = t + c$

$$\Rightarrow dx = dt$$

And at $x = a + c \Rightarrow t = a$

$\Rightarrow$ at $x = b + c \Rightarrow t = b$

Therefore,

$$\Rightarrow I = \int_a^b f(t+c) dt$$

$$\Rightarrow I = \int_a^b f(x+c) dx$$

Hence, option (B) is the correct answer.

24. If f and g are continuous function in $[0,1]$ satisfying $f(x) = f(a-x)$ and $g(x) + g(a-x) = a$, then $\int_0^a f(x) \cdot g(x) dx$ is equal to

(A) $\frac{a}{2}$

(B) $\frac{a}{2} \int_0^a f(x) dx$

(C) $\int_0^a f(x) dx$

(D) $a \int_0^a f(x) dx$

Ans: Let $I = \int_0^a f(x) \cdot g(x) dx$

$$\Rightarrow I = \int_0^a f(x-a) \cdot g(x-a) dx$$

$$\Rightarrow I = \int_0^a f(x) \cdot (a - g(x)) dx$$

$$\Rightarrow I = \int_0^a [a f(x) - f(x) \cdot g(x)] dx$$

$$\Rightarrow I = a \int_0^a f(x) dx - \int_0^a f(x) \cdot g(x) dx$$

$$\Rightarrow I = a \int_0^a f(x) dx - I$$

$$\Rightarrow 2I = a \int_0^a f(x) dx$$

$$\Rightarrow I = \frac{a}{2} \int_0^a f(x) dx$$

Hence, option (B) is the correct answer.

25: If $x = \int_0^y \frac{dt}{\sqrt{1+9t^2}}$ and $\frac{d^2y}{dx^2} = ay$, then a is equal to

(A) 3

(B) 6

(C) 9

(D) 1

Ans: here, we have $x = \int_0^y \frac{dt}{\sqrt{1+9t^2}}$

Differentiate w. r.t y , we get

$$\Rightarrow \frac{dx}{dy} = \frac{1}{\sqrt{1+9y^2}}$$

$$\Rightarrow \frac{dy}{dx} = \sqrt{1+9y^2}$$

Now, differentiate w. r.t x , we get

$$\Rightarrow \frac{d^2y}{dx^2} = \frac{18y}{2\sqrt{1+9y^2}} \cdot \frac{dy}{dx}$$

$$\Rightarrow \frac{d^2y}{dx^2} = \frac{9y}{\sqrt{1+9y^2}} \cdot \sqrt{1+9y^2}$$

$$\Rightarrow \frac{d^2y}{dx^2} = 9y$$

Thus, value of x is 9.

Hence, option (C) is the correct answer.

26: $\int_{-1}^1 \frac{x^3 + |x| + 1}{x^2 + 2|x| + 1} dx$ is equal to

(A) $\log 2$

(B) $2\log 2$

(C) $\frac{1}{2}\log 2$

(D) $4\log 2$

Ans: Let $I = \int_{-1}^1 \frac{x^3 + |x| + 1}{x^2 + 2|x| + 1} dx$

$$\Rightarrow I = \int_{-1}^1 \frac{x^3}{x^2 + 2|x| + 1} dx + \int_{-1}^1 \frac{|x| + 1}{x^2 + 2|x| + 1} dx$$

Here, $\int_{-1}^1 \frac{x^3}{x^2 + 2|x| + 1} dx$ is odd function and $\int_{-1}^1 \frac{|x| + 1}{x^2 + 2|x| + 1} dx$ is even function.

$$\Rightarrow I = 0 + 2 \int_0^1 \frac{|x| + 1}{(|x| + 1)^2} dx$$

$$\Rightarrow I = 2 \int_0^1 \frac{x + 1}{(x + 1)^2} dx$$

$$\Rightarrow I = 2 \int_0^1 \frac{1}{(x + 1)} dx$$

$$\Rightarrow I = 2 [\log |x + 1|]_0^1$$

$$\Rightarrow I = 2 [\log 2 - \log 1]$$

$$\Rightarrow I = 2 \log 2$$

Hence, option (B) is the correct answer.

27. If $\int_0^1 \frac{e^t}{1+t} dt = a$, then $\int_0^1 \frac{e^t}{(1+t)^2} dt$ is equal to

(A) $a - 1 + \frac{e}{2}$

(B) $a + 1 - \frac{e}{2}$

(C) $a - 1 - \frac{e}{2}$

(D) $a + 1 + \frac{e}{2}$

Ans: here, we have $\int_0^1 \frac{e^t}{1+t} dt = a$,

$$\Rightarrow a = \int_0^1 \frac{e^t}{1+t} dt$$

Applying integrating by parts, we get

$$\Rightarrow a = \left[e^t \times \frac{1}{(1+t)} \right]_0^1 + \int_0^1 \frac{e^t}{(1+t)^2} dt$$

$$\Rightarrow a = \left[\frac{e}{2} - 1 \right] + \int_0^1 \frac{e^t}{(1+t)^2} dt$$

$$\Rightarrow \int_0^1 \frac{e^t}{(1+t)^2} dt = a + 1 - \frac{e}{2}$$

Hence, option (B) is the correct answer.

28. $\int_{-2}^2 |x \cos \pi x| dx$ is equal to

(A) $\frac{8}{\pi}$

(B) $\frac{4}{\pi}$

(C) $\frac{2}{\pi}$

(D) $\frac{1}{\pi}$

Ans: Let $I = \int_{-2}^2 |x \cos \pi x| dx$

Here, it is a even function, therefore

$$\Rightarrow I = 2 \int_0^2 |x \cos \pi x| dx$$

$$\Rightarrow \frac{I}{2} = \int_0^{\frac{1}{2}} x \cos \pi x dx + \int_{\frac{1}{2}}^{\frac{3}{2}} -x \cos \pi x dx + \int_{\frac{3}{2}}^2 x \cos \pi x dx$$

$$\Rightarrow \frac{I}{2} = \left[\frac{x}{\pi} \sin \pi x + \frac{1}{\pi^2} \cos \pi x \right]_0^{\frac{1}{2}} - \left[\frac{x}{\pi} \sin \pi x + \frac{1}{\pi^2} \cos \pi x \right]_{\frac{1}{2}}^{\frac{3}{2}} + \left[\frac{x}{\pi} \sin \pi x + \frac{1}{\pi^2} \cos \pi x \right]_{\frac{3}{2}}^2$$

$$\Rightarrow \frac{I}{2} = \left[\frac{1}{2\pi} - \frac{1}{\pi^2} \right] - \left[-\frac{3}{2\pi} - \frac{1}{2\pi} \right] + \left[\frac{1}{\pi^2} + \frac{3}{2\pi} \right]$$

$$\Rightarrow \frac{I}{2} = \frac{1}{2\pi} - \frac{1}{\pi^2} + \frac{3}{2\pi} + \frac{1}{2\pi} + \frac{1}{\pi^2} + \frac{3}{2\pi}$$

$$\Rightarrow \frac{I}{2} = \frac{8}{2\pi}$$

$$\Rightarrow I = \frac{8}{\pi}$$

Hence, option (A) is the correct answer.

Fill in the blanks in each of the 29 to 32.

29. $\int \frac{\sin^6 x}{\cos^8 x} dx = \dots\dots\dots$

Ans: Let $I = \int \frac{\sin^6 x}{\cos^8 x} dx$

$\Rightarrow I = \int \frac{\sin^6 x}{\cos^6 x \cdot \cos^2 x} dx$

$\Rightarrow I = \int \tan^6 x \cdot \sec^2 x dx$

Put $\tan x = t$

$\Rightarrow \sec^2 x dx = dt$

Therefore,

$\Rightarrow I = \int t^6 dt$

$\Rightarrow I = \frac{t^7}{7} + C$

$\Rightarrow I = \frac{\tan^7 x}{7} + C$

Hence, $\int \frac{\sin^6 x}{\cos^8 x} dx = \frac{\tan^7 x}{7} + C$.

30. $\int_{-a}^a f(x) dx = 0$ if f is an function.

Ans: As we know that by the property of integration, $\int_{-a}^a f(x) dx = 0$ if f is an **odd** function.

31. $\int_0^{2a} f(x) dx = 2 \int_0^a f(x) dx$, if $f(2a-x) = \dots\dots\dots$

Ans: As, we know that by the property of integration, we have

$$\Rightarrow \int_0^{2a} f(x) dx = 2 \int_0^a f(x) dx, \text{ if } f(2a-x) = f(x)$$

$$= 0, \text{ if } f(2a-x) = -f(x)$$

Hence, $\int_0^{2a} f(x) dx = 2 \int_0^a f(x) dx$, if $f(2a-x) = f(x)$.

32. $\int_0^{\frac{\pi}{2}} \frac{\sin^n x}{\sin^n x + \cos^n x} dx = \dots\dots\dots$

Ans: Let $I = \int_0^{\frac{\pi}{2}} \frac{\sin^n x}{\sin^n x + \cos^n x} dx \dots\dots\dots (i)$

$$I = \int_0^{\frac{\pi}{2}} \frac{\sin^n \left(\frac{\pi}{2} - x \right)}{\sin^n \left(\frac{\pi}{2} - x \right) + \cos^n \left(\frac{\pi}{2} - x \right)} dx$$

$$\Rightarrow$$

$$\Rightarrow I = \int_0^{\frac{\pi}{2}} \frac{\cos^n x}{\cos^n x + \sin^n x} dx \dots\dots\dots (ii)$$

Adding eq (i) and (ii), we get

$$\Rightarrow 2I = \int_0^{\frac{\pi}{2}} dx$$

$$\Rightarrow 2I = [x]_0^{\frac{\pi}{2}}$$

$$\Rightarrow 2I = \frac{\pi}{2}$$

$$\Rightarrow I = \frac{\pi}{4}$$

Hence, $\int_0^{\frac{\pi}{2}} \frac{\sin^n x}{\sin^n x + \cos^n x} dx = \frac{\pi}{4}.$

Exercise 7.3

Short Answer (S.A)

Verify the following:

1. $\int \frac{2x-1}{2x+3} dx = x - \log |(2x+3)| + C$

Ans: Let $I = \int \frac{2x-1}{2x+3} dx$

$$\Rightarrow I = \int \frac{2x+3-3-1}{2x+3} dx$$

$$\Rightarrow I = \int \frac{2x+3-4}{2x+3} dx$$

$$\Rightarrow I = \int \left(1 - \frac{1}{2x+3} \right) dx$$

$$\Rightarrow I = \int dx - \int \frac{1}{2x+3} dx$$

$$\Rightarrow I = x - \int \frac{1}{2\left(x + \frac{3}{2}\right)} dx$$

$$\Rightarrow I = x - 2 \log \left(x + \frac{3}{2} \right) + C$$

$$\Rightarrow I = x - 2 \log \left(\frac{2x+3}{2} \right) + C$$

$$\Rightarrow I = x - 2[\log(2x+3) - \log 2] + C$$

$$\Rightarrow I = x - 2 \log(2x+3) + 2\log 2 + C$$

$$\Rightarrow I = x - \log |(2x+3)^2| + C$$

Thus, $\int \frac{2x-1}{2x+3} dx = x - \log |(2x-3)^2| + C$

Hence proved.

2. $\int \frac{2x+3}{x^2+3x} dx = \log |x^2+3x| + C$

Ans: let $I = \int \frac{2x+3}{x^2+3x} dx$

Put $x^2+3x = t$

$$\Rightarrow (2x+3)dx = dt$$

$$\therefore I = \int \frac{1}{t} dt$$

$$\Rightarrow I = \log |t| + C$$

$$\Rightarrow I = \log |(x^2+3x)| + C$$

Thus, $\int \frac{2x+3}{x^2+3x} dx = \log |x^2+3x| + C$

Hence proved.

Evaluate the following:

3. $\int \frac{(x^2+2)dx}{x+1}$

Ans: Let $I = \int \frac{(x^2 + 2)dx}{x+1}$

$$\Rightarrow I = \int \left(x - 1 + \frac{3}{x+1} \right) dx$$

$$\Rightarrow I = \int x dx - \int dx + \int \frac{3}{x+1} dx$$

$$\Rightarrow I = \frac{x^2}{2} - x + 3\log|x+1| + C$$

Hence, $\int \frac{(x^2 + 2)dx}{x+1} = \frac{x^2}{2} - x + 3\log|x+1| + C.$

4. $\int \frac{e^{6\log x} - e^{5\log x}}{e^{4\log x} - e^{3\log x}} dx$

Ans: Let $I = \int \frac{e^{6\log x} - e^{5\log x}}{e^{4\log x} - e^{3\log x}} dx$

$$\Rightarrow I = \int \frac{e^{\log x^6} - e^{\log x^5}}{e^{\log x^4} - e^{\log x^3}} dx$$

$$\Rightarrow I = \int \frac{x^6 - x^5}{x^4 - x^3} dx \quad [e^{\log x} = x]$$

$$\Rightarrow I = \int \frac{x^5(x-1)}{x^3(x-1)} dx$$

$$\Rightarrow I = \int x^2 dx$$

$$\Rightarrow I = \frac{x^3}{3} + C$$

Hence, $\int \frac{e^{6\log x} - e^{5\log x}}{e^{4\log x} - e^{3\log x}} dx = \frac{x^3}{3} + C.$

5. $\int \frac{(1 + \cos x)}{x + \sin x} dx$

Ans: Let $I = \int \frac{(1 + \cos x)}{x + \sin x} dx$

Put $x + \sin x = t$

$\Rightarrow (1 + \cos x) dx = dt$

$\Rightarrow I = \int \frac{1}{t} dx$

$\Rightarrow I = \log|t| + C$

$\Rightarrow I = \log|(x + \sin x)| + C$

6. $\int \frac{dx}{1 + \cos x}$

Ans: Let $I = \int \frac{dx}{1 + \cos x}$

$\Rightarrow I = \int \frac{dx}{2 \cos^2 \frac{x}{2}}$

$\Rightarrow I = \frac{1}{2} \int \frac{dx}{\cos^2 \frac{x}{2}}$

$\Rightarrow I = \frac{1}{2} \int \sec^2 \frac{x}{2} dx$

$\Rightarrow I = \frac{1}{2} \times 2 \tan \frac{x}{2} + C$ $[\int \sec^2 x dx = \tan x + C]$

$$\Rightarrow I = \tan \frac{x}{2} + C$$

Hence, $\int \frac{dx}{1 + \cos x} = \tan \frac{x}{2} + C$.

7. $\int \tan^2 x \cdot \sec^4 x \, dx$

Ans: Let $I = \int \tan^2 x \cdot \sec^4 x \, dx$

$$\Rightarrow I = \int \tan^2 x \cdot \sec^2 x \cdot \sec^2 x \, dx$$

$$\Rightarrow I = \int \tan^2 x (1 + \tan^2 x) \sec^2 x \, dx$$

Put $\tan x = t$

$$\Rightarrow \sec^2 x \, dx = dt$$

Therefore,

$$\Rightarrow I = \int t^2 (1 + t^2) \, dt$$

$$\Rightarrow I = \int (t^2 + t^4) \, dt$$

$$\Rightarrow I = \frac{t^3}{3} + \frac{t^5}{5} + C$$

$$\Rightarrow I = \frac{\tan^3 x}{3} + \frac{\tan^5 x}{5} + C$$

8. $\int \frac{\sin x + \cos x}{\sqrt{1 + \sin 2x}} \, dx$

Ans: Let $I = \int \frac{\sin x + \cos x}{\sqrt{1 + \sin 2x}} \, dx$

$$\Rightarrow I = \int \frac{\sin x + \cos x}{\sqrt{\sin^2 x + \cos^2 x + 2 \sin x \cos x}} dx$$

$$\Rightarrow I = \int \frac{\sin x + \cos x}{\sqrt{(\sin x + \cos x)^2}} dx$$

$$\Rightarrow I = \int \frac{(\sin x + \cos x)}{(\sin x + \cos x)} dx$$

$$\Rightarrow I = \int dx$$

$$\Rightarrow I = x + C$$

Hence, $\int \frac{\sin x + \cos x}{\sqrt{1 + \sin 2x}} dx = x + C$.

9. $\int \sqrt{1 + \sin x} dx$

Ans: Let $I = \int \sqrt{1 + \sin x} dx$

$$\Rightarrow I = \int \sqrt{\sin^2 \frac{x}{2} + \cos^2 \frac{x}{2} + 2 \sin \frac{x}{2} \cos \frac{x}{2}} dx$$

$$\Rightarrow I = \int \sqrt{\left(\sin \frac{x}{2} + \cos \frac{x}{2}\right)^2} dx$$

$$\Rightarrow I = \int \left(\sin \frac{x}{2} + \cos \frac{x}{2}\right) dx$$

$$\Rightarrow I = \int \sin \frac{x}{2} dx + \int \cos \frac{x}{2} dx$$

$$\Rightarrow I = -2 \cos \frac{x}{2} + 2 \sin \frac{x}{2} + C$$

Hence, $\int \sqrt{1 + \sin x} dx = -2 \cos \frac{x}{2} + 2 \sin \frac{x}{2} + C$.

10. $\int \frac{x}{\sqrt{x}+1} dx$

Ans: Let $I = \int \frac{x}{\sqrt{x}+1} dx$

Put $\sqrt{x} = z$

$$\Rightarrow \frac{1}{2\sqrt{x}} dx = dz$$

$$\Rightarrow dx = 2\sqrt{x} dz$$

$$\Rightarrow dx = 2z dz$$

Therefore,

$$\Rightarrow I = \int \frac{z^2}{z+1} \times 2z dz$$

$$\Rightarrow I = 2 \int \frac{z^3}{z+1} dz$$

$$\Rightarrow I = 2 \int \frac{z^3 + 1 - 1}{z+1} dz$$

$$\Rightarrow I = 2 \int \frac{z^3 + 1}{z+1} dz - 2 \int \frac{1}{z+1} dz$$

$$\Rightarrow I = 2 \int \frac{(z+1)(z^2 - z + 1)}{z+1} dz - 2 \int \frac{1}{z+1} dz$$

$$\Rightarrow I = 2 \int (z^2 - z + 1) dz - 2 \int \frac{1}{z+1} dz$$

$$\Rightarrow I = 2 \left(\frac{z^3}{3} - \frac{z^2}{2} + z \right) - 2 \log|z+1| + C$$

$$\Rightarrow I = 2 \left(\frac{x\sqrt{x}}{3} - \frac{x}{2} + \sqrt{x} \right) - 2 \log |\sqrt{x} + 1| + C$$

$$\Rightarrow I = 2 \left(\frac{x\sqrt{x}}{3} - \frac{x}{2} + \sqrt{x} - \log |\sqrt{x} + 1| \right) + C$$

Hence, $\int \frac{x}{\sqrt{x} + 1} dx = 2 \left(\frac{x\sqrt{x}}{3} - \frac{x}{2} + \sqrt{x} - \log |\sqrt{x} + 1| \right) + C .$

11. $\int \sqrt{\frac{a+x}{a-x}} dx$

Ans: Let $I = \int \sqrt{\frac{a+x}{a-x}} dx$

$$\Rightarrow I = \int \sqrt{\frac{(a+x)(a+x)}{(a-x)(a+x)}} dx$$

$$\Rightarrow I = \int \sqrt{\frac{(a+x)^2}{(a^2 - x^2)}} dx$$

$$\Rightarrow I = \int \frac{a+x}{\sqrt{(a^2 - x^2)}} dx$$

$$\Rightarrow I = \int \frac{a}{\sqrt{(a^2 - x^2)}} dx + \int \frac{x}{\sqrt{(a^2 - x^2)}} dx$$

$$\Rightarrow I = a \sin^{-1} \left(\frac{x}{a} \right) + \int \frac{x}{\sqrt{(a^2 - x^2)}} dx$$

Put $a^2 - x^2 = t^2$

$$\Rightarrow -2x dx = 2t dt$$

$$\Rightarrow x dx = -t dt$$

Therefore,

$$\Rightarrow I = a \sin^{-1}\left(\frac{x}{a}\right) + \int \frac{-t dt}{\sqrt{t^2}}$$

$$\Rightarrow I = a \sin^{-1}\left(\frac{x}{a}\right) - \int \frac{t dt}{t}$$

$$\Rightarrow I = a \sin^{-1}\left(\frac{x}{a}\right) - \int dt$$

$$\Rightarrow I = a \sin^{-1}\left(\frac{x}{a}\right) - t + C$$

$$\Rightarrow I = a \sin^{-1}\left(\frac{x}{a}\right) - \sqrt{(a^2 - x^2)} + C$$

$$12. \int \frac{x^{\frac{1}{2}}}{1+x^{\frac{3}{4}}} dx$$

$$I = \int \frac{x^{\frac{1}{2}}}{1+x^{\frac{3}{4}}} dx$$

Ans: Let

$$\text{Put } x = z^4$$

$$\Rightarrow dx = 4z^3 dz$$

Therefore,

$$\Rightarrow I = \int \frac{(z^4)^{\frac{1}{2}}}{1+(z^4)^{\frac{3}{4}}} \times 4z^3 dz$$

$$\Rightarrow I = 4 \int \frac{z^2 \cdot z^3}{1+z^3} dz$$

Now, put $1+z^3 = t$

$$\Rightarrow 3z^2 dz = dt$$

$$\Rightarrow z^2 dz = \frac{dt}{3}$$

Therefore, we get

$$\Rightarrow I = 4 \int \frac{(t-1)}{t} \cdot \frac{dt}{3}$$

$$\Rightarrow I = \frac{4}{3} \int \left(1 - \frac{1}{t}\right) dt$$

$$\Rightarrow I = \frac{4}{3} t - \frac{4}{3} \log|t| + C_1$$

$$\Rightarrow I = \frac{4}{3} (1 + z^3) - \frac{4}{3} \log|1 + z^3| + C_1$$

$$\Rightarrow I = \frac{4}{3} \left(1 + x^{\frac{3}{4}}\right) - \frac{4}{3} \log\left|1 + x^{\frac{3}{4}}\right| + C_1$$

$$\Rightarrow I = \frac{4}{3} + \frac{4}{3} x^{\frac{3}{4}} - \frac{4}{3} \log\left|1 + x^{\frac{3}{4}}\right| + C_1$$

$$\Rightarrow I = \frac{4}{3} x^{\frac{3}{4}} - \frac{4}{3} \log\left|1 + x^{\frac{3}{4}}\right| + C$$

13. $\int \frac{\sqrt{1+x^2}}{x^4} dx$

Ans: Let $I = \int \frac{\sqrt{1+x^2}}{x^4} dx$

$$\Rightarrow I = \int \frac{\sqrt{1+x^2}}{x \cdot x^3} dx$$

$$\Rightarrow I = \int \sqrt{\frac{1+x^2}{x^2}} \frac{dx}{x^3}$$

$$\Rightarrow I = \int \sqrt{\left(\frac{1}{x^2} + 1\right)} \frac{dx}{x^3}$$

$$\text{Put } \left(\frac{1}{x^2} + 1\right) = t^2$$

$$\Rightarrow -\frac{2}{x^3} dx = 2t dt$$

$$\Rightarrow \frac{dx}{x^3} = -t dt$$

Therefore,

$$\Rightarrow I = \int \sqrt{t^2} (-t) dt$$

$$\Rightarrow I = -\int t \cdot t dt$$

$$\Rightarrow I = -\int t^2 dt$$

$$\Rightarrow I = -\frac{t^3}{3} + C$$

$$\Rightarrow I = -\frac{1}{3} \left(\frac{1}{x^2} + 1\right)^{\frac{3}{2}} + C$$

14. $\int \frac{dx}{\sqrt{16-9x^2}}$

Ans: Let $I = \int \frac{dx}{\sqrt{16-9x^2}}$

$$I = \int \frac{dx}{\sqrt{9\left(\frac{16}{9} - x^2\right)}}$$

$$\Rightarrow$$

$$I = \frac{1}{3} \int \frac{dx}{\sqrt{\left(\frac{4}{3}\right)^2 - x^2}}$$

$$\Rightarrow$$

$$I = \frac{1}{3} \sin^{-1} \left(\frac{x}{\frac{4}{3}} \right) + C$$

$$\Rightarrow$$

$$\Rightarrow I = \frac{1}{3} \sin^{-1} \left(\frac{3x}{4} \right) + C$$

15. $\int \frac{dt}{\sqrt{3t - 2t^2}}$

Ans: Let $I = \int \frac{dt}{\sqrt{3t - 2t^2}}$

$$I = \int \frac{dt}{\sqrt{-2\left(t^2 - \frac{3t}{2}\right)}}$$

$$\Rightarrow$$

$$I = \frac{1}{\sqrt{2}} \int \frac{dt}{\sqrt{-\left(t^2 - \frac{3t}{2}\right)}}$$

$$\Rightarrow$$

$$I = \frac{1}{\sqrt{2}} \int \frac{dt}{\sqrt{-\left[t^2 - \frac{3t}{2} + \left(\frac{3}{4}\right)^2 - \left(\frac{3}{4}\right)^2\right]}}$$

$$\Rightarrow$$

$$I = \frac{1}{\sqrt{2}} \int \frac{dt}{\sqrt{-\left[t - \frac{3}{4}\right]^2 - \left(\frac{3}{4}\right)^2}}$$

$$\Rightarrow$$

$$I = \frac{1}{\sqrt{2}} \int \frac{dt}{\sqrt{\left(\frac{3}{4}\right)^2 - \left(t - \frac{3}{4}\right)^2}}$$

$$\Rightarrow$$

$$I = \frac{1}{\sqrt{2}} \sin^{-1} \left(\frac{t - \frac{3}{4}}{\frac{3}{4}} \right) + C$$

$$\Rightarrow$$

$$\Rightarrow I = \frac{1}{\sqrt{2}} \sin^{-1} \left(\frac{4t - 3}{3} \right) + C$$

16. $\int \frac{3x-1}{\sqrt{x^2+9}} dx$

Ans: Let $I = \int \frac{3x-1}{\sqrt{x^2+9}} dx$

$$\Rightarrow I = \int \frac{3x}{\sqrt{x^2+9}} dx - \int \frac{dx}{\sqrt{x^2+9}}$$

$$\Rightarrow I = I_1 - I_2 \dots\dots\dots (i),$$

Where, $I_1 = \int \frac{3x}{\sqrt{x^2+9}} dx$ and $I_2 = \int \frac{dx}{\sqrt{x^2+9}}$

Therefore,

$$\Rightarrow I_1 = \int \frac{3x}{\sqrt{x^2+9}} dx$$

Put $x^2 + 9 = t^2$

$$\Rightarrow 2x dx = 2t dt$$

$$\Rightarrow x dx = t dt$$

$$\Rightarrow I_1 = \int \frac{3t dt}{\sqrt{t^2}}$$

$$\Rightarrow I_1 = \int \frac{3t dt}{t}$$

$$\Rightarrow I_1 = \int 3 dt$$

$$\Rightarrow I_1 = 3t + C_1$$

$$\Rightarrow I_1 = 3\sqrt{x^2 + 9} + C_1 \dots\dots\dots (ii)$$

Also we have,

$$\Rightarrow I_2 = \int \frac{dx}{\sqrt{x^2 + 9}}$$

$$\Rightarrow I_2 = \int \frac{dx}{\sqrt{x^2 + 3^2}}$$

$$\Rightarrow I_2 = \log|x + \sqrt{x^2 + 9}| + C_2 \dots\dots\dots (iii)$$

Put the value of I_1 and I_2 in eq (i), we get

$$\Rightarrow I = 3\sqrt{x^2 + 9} + C_1 - \log|x + \sqrt{x^2 + 9}| - C_2$$

$$\Rightarrow I = 3\sqrt{x^2 + 9} - \log|x + \sqrt{x^2 + 9}| + C$$

17. $\int \sqrt{5 - 2x + x^2} dx$

Ans: Let $I = \int \sqrt{5 - 2x + x^2} dx$

$$\Rightarrow I = \int \sqrt{x^2 - 2x + 5} dx.$$

$$\Rightarrow I = \int \sqrt{x^2 - 2x + 1 + 4} dx$$

$$\Rightarrow I = \int \sqrt{(x^2 - 2x + 1) + 2^2} dx$$

$$\Rightarrow I = \int \sqrt{(x-1)^2 + 2^2} dx$$

$$\Rightarrow I = \frac{x-1}{2} \sqrt{(x-1)^2 + 2^2} + \frac{2^2}{2} \log \left| x-1 + \sqrt{(x-1)^2 + 2^2} \right| + C$$

$$\Rightarrow I = \frac{(x-1)}{2} \sqrt{5-2x+x^2} + 2 \log \left| x-1 + \sqrt{5-2x+x^2} \right| + C.$$

18. $\int \frac{x}{x^4-1} dx$.

Ans: Let $I = \int \frac{x}{x^4-1} dx$

$$\Rightarrow I = \int \frac{x}{(x^2)^2-1} dx$$

Put $x^2 = t$.

$$\Rightarrow 2x dx = dt.$$

$$\Rightarrow x dx = \frac{dt}{2}$$

Therefore,

$$\Rightarrow I = \frac{1}{2} \int \frac{dt}{(t)^2-1}$$

$$\Rightarrow I = \frac{1}{2} \cdot \frac{1}{2 \times 1} \log \left| \frac{t-1}{t+1} \right| + C$$

$$\Rightarrow I = \frac{1}{4} \log \left| \frac{x^2-1}{x^2+1} \right| + C$$

19. $\int \frac{x^2}{1-x^4} dx$

Ans: Let $I = \int \frac{x^2}{1-x^4} dx$

$$\Rightarrow I = \int \frac{\frac{x^2}{2} + \frac{x^2}{2}}{1^2 - (x^2)^2} dx$$

$$\Rightarrow I = \int \frac{\frac{1}{2} + \frac{x^2}{2} - \frac{1}{2} + \frac{x^2}{2}}{(1-x^2)(1+x^2)} dx$$

$$\Rightarrow I = \int \frac{\frac{1}{2}(1+x^2) - \frac{1}{2}(1-x^2)}{(1-x^2)(1+x^2)} dx$$

$$\Rightarrow I = \frac{1}{2} \int \frac{(1+x^2)}{(1-x^2)(1+x^2)} dx - \frac{1}{2} \int \frac{(1-x^2)}{(1-x^2)(1+x^2)} dx$$

$$\Rightarrow I = \frac{1}{2} \int \frac{dx}{(1-x^2)} - \frac{1}{2} \int \frac{dx}{(1+x^2)}$$

$$\Rightarrow I = \frac{1}{2} \cdot \frac{1}{2 \times 1} \log \left| \frac{1+x}{1-x} \right| - \frac{1}{2} \tan^{-1} x + C$$

$$\Rightarrow I = \frac{1}{4} \log \left| \frac{1+x}{1-x} \right| - \frac{1}{2} \tan^{-1} x + C$$

20. $\int \sqrt{2ax - x^2} dx$

Ans: Let $I = \int \sqrt{2ax - x^2} dx$

$$\Rightarrow I = \int \sqrt{-(x^2 - 2ax)} dx$$

$$\Rightarrow I = \int \sqrt{-(x^2 - 2ax + a^2 - a^2)} dx$$

$$\Rightarrow I = \int \sqrt{-(x^2 - 2ax + a^2) - a^2} dx$$

$$\Rightarrow I = \int \sqrt{-(x-a)^2 - a^2} dx$$

$$\Rightarrow I = \int \sqrt{a^2 - (x-a)^2} dx$$

$$\Rightarrow I = \frac{(x-a)}{2} \sqrt{a^2 - (x-a)^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x-a}{a} \right) + C$$

$$\Rightarrow I = \frac{(x-a)}{2} \sqrt{2ax - x^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x-a}{a} \right) + C$$

21. $\int \frac{\sin^{-1} x}{(1-x^2)^{\frac{3}{2}}} dx$

Let $I = \int \frac{\sin^{-1} x}{(1-x^2)^{\frac{3}{2}}} dx$

$$\Rightarrow I = \int \frac{\sin^{-1} x}{(1-x^2)\sqrt{1-x^2}} dx$$

Put $\sin^{-1} x = t$,

$$\Rightarrow \frac{1}{\sqrt{1-x^2}} dx = dt$$

And $\sin^{-1} x = t \Rightarrow x = \sin t$

$$\cos t = \sqrt{1-x^2}$$

Therefore,

$$I = \int \frac{t \cdot dt}{(1 - \sin^2 t)}$$

$$\Rightarrow I = \int \frac{t \cdot dt}{\cos^2 t}$$

$$\Rightarrow I = \int t \cdot \sec^2 t \, dt$$

$$\Rightarrow I = t \int \sec^2 t \, dt - \int \left(\frac{d}{dt} t \cdot \int \sec^2 t \, dt \right) dt$$

$$\Rightarrow I = t \tan t - \int \tan t \, dt$$

$$\Rightarrow I = t \tan t + \log |\cos t| + C$$

$$\Rightarrow I = \sin^{-1} x \times \frac{\sin t}{\cos t} + \log |\sqrt{1 - x^2}| + C$$

$$\Rightarrow I = \frac{x \sin^{-1} x}{\sqrt{1 - x^2}} + \log |\sqrt{1 - x^2}| + C$$

$$\text{Hence, } \int \frac{\sin^{-1} x}{(1 - x^2)^{\frac{3}{2}}} dx = \frac{x \sin^{-1} x}{\sqrt{1 - x^2}} + \log |\sqrt{1 - x^2}| + C.$$

22. $\int \frac{(\cos 5x + \cos 4x)}{1 - 2 \cos 3x} dx$

Ans: Let $I = \int \frac{(\cos 5x + \cos 4x)}{1 - 2 \cos 3x} dx$.

Applying $[\cos C + \cos D = 2 \cos \frac{C+D}{2} \cdot \cos \frac{C-D}{2}]$ and $\cos 2x = 2 \cos^2 x - 1$

$$\Rightarrow I = \int \frac{2 \cos \frac{9x}{2} \cdot \cos \frac{x}{2}}{1 - 2 \left(2 \cos^2 \frac{3x}{2} - 1 \right)} dx$$

$$I = \int \frac{2 \cos \frac{9x}{2} \cdot \cos \frac{x}{2}}{1 - 4 \cos^2 \frac{3x}{2} + 2} dx$$

$$\Rightarrow$$

$$I = \int \frac{2 \cos \frac{9x}{2} \cdot \cos \frac{x}{2}}{3 - 4 \cos^2 \frac{3x}{2}} dx$$

$$\Rightarrow$$

Now, multiply and divide by $\cos \frac{3x}{2}$, we get

$$I = \int \frac{2 \cos \frac{9x}{2} \cdot \cos \frac{x}{2} \cdot \cos \frac{3x}{2}}{3 \cos \frac{3x}{2} - 4 \cos^3 \frac{3x}{2}} dx$$

$$\Rightarrow$$

$$I = - \int \frac{2 \cos \frac{9x}{2} \cdot \cos \frac{x}{2} \cdot \cos \frac{3x}{2}}{4 \cos^3 \frac{3x}{2} - 3 \cos \frac{3x}{2}} dx$$

$$\Rightarrow$$

$$I = - \int \frac{2 \cos \frac{9x}{2} \cdot \cos \frac{x}{2} \cdot \cos \frac{3x}{2}}{\cos 3 \cdot \frac{3x}{2}} dx$$

$$\Rightarrow$$

$$I = - \int \frac{2 \cos \frac{9x}{2} \cdot \cos \frac{x}{2} \cdot \cos \frac{3x}{2}}{\cos \frac{9x}{2}} dx$$

$$\Rightarrow$$

$$\Rightarrow I = - \int 2 \cos \frac{x}{2} \cdot \cos \frac{3x}{2} dx$$

$$\Rightarrow$$

$$\Rightarrow I = - \int \left[\cos \left(\frac{3x}{2} + \frac{x}{2} \right) + \cos \left(\frac{3x}{2} - \frac{x}{2} \right) \right] dx$$

$$\Rightarrow$$

$$\Rightarrow I = - \int (\cos 2x + \cos x) dx$$

$$\Rightarrow I = -\int \cos 2x dx - \int \cos x dx$$

$$\Rightarrow I = -\left(\frac{\sin 2x}{2}\right) - \sin x + C$$

Hence,
$$\int \frac{(\cos 5x + \cos 4x)}{1 - 2\cos 3x} dx = -\left(\frac{\sin 2x}{2}\right) - \sin x + C.$$

23.
$$\int \frac{\sin^6 x + \cos^6 x}{\sin^2 x \cdot \cos^2 x} dx$$

Ans: Let
$$I = \int \frac{\sin^6 x + \cos^6 x}{\sin^2 x \cdot \cos^2 x} dx$$

$$\Rightarrow I = \int \frac{(\sin^2 x)^3 + (\cos^2 x)^3}{\sin^2 x \cdot \cos^2 x} dx$$

$$\Rightarrow I = \int \frac{(\sin^2 x + \cos^2 x)(\sin^4 x - \sin^2 x \cdot \cos^2 x + \cos^4 x)}{\sin^2 x \cdot \cos^2 x} dx$$

$$\Rightarrow I = \int \frac{(\sin^4 x - \sin^2 x \cdot \cos^2 x + \cos^4 x)}{\sin^2 x \cdot \cos^2 x} dx$$

$$\Rightarrow I = \int \frac{\sin^4 x}{\sin^2 x \cdot \cos^2 x} dx - \int \frac{\sin^2 x \cdot \cos^2 x}{\sin^2 x \cdot \cos^2 x} dx + \int \frac{\cos^4 x}{\sin^2 x \cdot \cos^2 x} dx$$

$$\Rightarrow I = \int \frac{\sin^2 x}{\cos^2 x} dx - \int dx + \int \frac{\cos^2 x}{\sin^2 x} dx$$

$$\Rightarrow I = \int \tan^2 x dx - \int dx + \int \cot^2 x dx$$

$$\Rightarrow I = \int (\sec^2 x - 1) dx - \int dx + \int (\operatorname{cosec}^2 x - 1) dx$$

$$\Rightarrow I = \int (\sec^2 x dx + \operatorname{cosec}^2 x dx - 3) dx$$

$$\Rightarrow I = \tan x - \cot x - 3x + C$$

Hence, $\int \frac{\sin^6 x + \cos^6 x}{\sin^2 x \cdot \cos^2 x} dx = \tan x - \cot x - 3x + C.$

24. $\int \frac{\sqrt{x}}{\sqrt{a^3 - x^3}} dx$

Ans: Let $I = \int \frac{\sqrt{x}}{\sqrt{a^3 - x^3}} dx$

$$\Rightarrow I = \int \frac{x^{\frac{1}{2}}}{\sqrt{\left(a^{\frac{3}{2}}\right)^2 - \left(x^{\frac{3}{2}}\right)^2}} dx$$

Here, Put $x^{\frac{3}{2}} = t$

$$\Rightarrow \frac{3}{2} x^{\frac{1}{2}} dx = dt$$

$$\Rightarrow x^{\frac{1}{2}} dx = \frac{2}{3} dt$$

Therefore,

$$\Rightarrow I = \frac{2}{3} \int \frac{dt}{\sqrt{\left(a^{\frac{3}{2}}\right)^2 - (t)^2}}$$

$$\Rightarrow I = \frac{2}{3} \sin^{-1} \left(\frac{t}{a^{\frac{3}{2}}} \right) + C$$

$$\Rightarrow I = \frac{2}{3} \sin^{-1} \left(\frac{x^{\frac{3}{2}}}{a^{\frac{3}{2}}} \right) + C$$

$$\Rightarrow I = \frac{2}{3} \sin^{-1} \left(\frac{x}{a} \right)^{\frac{3}{2}} + C$$

Hence, $\int \frac{\sqrt{x}}{\sqrt{a^3 - x^3}} dx = \frac{2}{3} \sin^{-1} \left(\frac{x}{a} \right)^{\frac{3}{2}} + C.$

25. $\int \frac{\cos x - \cos 2x}{1 - \cos x} dx$

Ans: Let $I = \int \frac{\cos x - \cos 2x}{1 - \cos x} dx$

$$\Rightarrow I = \int \frac{2 \sin \frac{3x}{2} \cdot \sin \frac{x}{2}}{2 \sin^2 \frac{x}{2}} dx$$

$$\Rightarrow I = \int \frac{\sin \frac{3x}{2}}{\sin \frac{x}{2}} dx$$

$$\Rightarrow I = \int \frac{3 \sin \frac{x}{2} - 4 \sin^3 \frac{x}{2}}{\sin \frac{x}{2}} dx$$

$$\Rightarrow I = 3 \int dx - 4 \int \sin^2 \frac{x}{2} dx$$

$$\Rightarrow I = 3x - 4 \int \frac{1 + \cos x}{2} dx$$

$$\Rightarrow I = 3x - 2 \int (1 + \cos x) dx$$

$$\Rightarrow I = 3x - 2(x + \sin x) + C$$

$$\Rightarrow I = 3x - 2x + 2 \sin x + C$$

$$\Rightarrow I = x + 2 \sin x + C$$

Hence, $\int \frac{\cos x - \cos 2x}{1 - \cos x} dx = x + 2 \sin x + C.$

26. $\int \frac{dx}{x\sqrt{x^4 - 1}}$

Ans: Let $I = \int \frac{dx}{x\sqrt{x^4 - 1}}$

Multiply and divide by x^3 , we get

$$\Rightarrow I = \int \frac{x^3 dx}{x^4 \sqrt{x^4 - 1}}$$

Now, put $x^4 - 1 = t^2$

$$\Rightarrow 4x^3 dx = 2t dt$$

$$\Rightarrow x^3 dx = \frac{t}{2} dt$$

Therefore,

$$\Rightarrow I = \frac{1}{2} \int \frac{t dt}{(1+t^2)t}$$

$$\Rightarrow I = \frac{1}{2} \int \frac{dt}{(1+t^2)}$$

$$\Rightarrow I = \frac{1}{2} \tan^{-1} t + C$$

$$\Rightarrow I = \frac{1}{2} \tan^{-1} (\sqrt{x^4 - 1}) + C$$

Hence, $\int \frac{dx}{x\sqrt{x^4 - 1}} = \frac{1}{2} \tan^{-1} (\sqrt{x^4 - 1}) + C.$

Evaluate the following as limit of sums:

$$27. \int_0^2 (x^2 + 3) dx$$

Ans: Let $I = \int_0^2 (x^2 + 3) dx$

Here, we have $a = 0, b = 2$ and $h = \frac{b-a}{n} = \frac{2}{n}$

$\Rightarrow nh = 2$ and $f(x) = (x^2 + 3)$

Now, $\int_0^2 (x^2 + 3) dx = \lim_{h \rightarrow 0} h [f(0) + f(0+h) + \dots + f\{0+(n-1)h\}]$

$$= \lim_{h \rightarrow 0} h [3 + h^2 + 3 + 2^2 h^2 + 3 + \dots + (n-1)^2 h^2 + 3]$$

$$= \lim_{h \rightarrow 0} h [3n + h^2 \{1^2 + 2^2 + 3^2 + \dots + (n-1)^2\}]$$

$$= \lim_{h \rightarrow 0} h \left[3n + h^2 \times \frac{n(n-1)(2n-1)}{6} \right]$$

$$= \lim_{h \rightarrow 0} \left[3nh + h^3 \times \frac{n(n-1)(2n-1)}{6} \right]$$

$$= \lim_{h \rightarrow 0} \left[3nh + \frac{nh(nh-h)(2nh-h)}{6} \right]$$

$$= \lim_{h \rightarrow 0} \left[3 \cdot 2 + \frac{2(2-h)(2 \cdot 2 - h)}{6} \right]$$

$$= 6 + \frac{4 \times 4}{6}$$

$$= 6 + \frac{16}{6}$$

$$= 6 + \frac{8}{3}$$

$$= \frac{26}{3}$$

Hence, $I = \int_0^2 (x^2 + 3) dx = \frac{26}{3}$

28. $\int_0^2 e^x dx$

Ans: let $I = \int_0^2 e^x dx$

Here, we have $a = 0, b = 2$ and $h = \frac{b-a}{n} = \frac{2}{n}$

$\Rightarrow nh = 2$ and $f(x) = e^x$

Now, $\int_0^2 e^x dx = \lim_{h \rightarrow 0} h \left[f(0) + f(0+h) + \dots + f\{0+(n-1)h\} \right]$

$$= \lim_{h \rightarrow 0} h \left[1 + e^h + e^{2h} + e^{3h} + \dots + e^{(n-1)h} \right]$$

$$= \lim_{h \rightarrow 0} h \left[\frac{1 \cdot (e^h)^n - 1}{e^h - 1} \right]$$

$$= \lim_{h \rightarrow 0} h \left[\frac{e^{nh} - 1}{e^h - 1} \right]$$

$$= \lim_{h \rightarrow 0} h \left[\frac{e^2 - 1}{e^h - 1} \right]$$

$$= e^2 \lim_{h \rightarrow 0} \frac{h}{e^h - 1} - \lim_{h \rightarrow 0} \frac{h}{e^h - 1}$$

$$= e^2 - 1$$

Hence, $I = \int_0^2 e^x dx = e^2 - 1$.

29. $\int_0^{\frac{\pi}{2}} \frac{\tan x dx}{1 + m^2 \tan^2 x}$

Ans: Let $I = \int_0^{\frac{\pi}{2}} \frac{\tan x dx}{1 + m^2 \tan^2 x}$

$$\Rightarrow I = \int_0^{\frac{\pi}{2}} \frac{\frac{\sin x}{\cos x} dx}{1 + m^2 \frac{\sin^2 x}{\cos^2 x}}$$

$$\Rightarrow I = \int_0^{\frac{\pi}{2}} \frac{\frac{\sin x}{\cos x} dx}{\cos^2 x + m^2 \sin^2 x}$$

$$\Rightarrow I = \int_0^{\frac{\pi}{2}} \frac{\sin x \cdot \cos x dx}{\cos^2 x + m^2 \sin^2 x}$$

$$\Rightarrow I = \int_0^{\frac{\pi}{2}} \frac{\sin x \cdot \cos x dx}{1 - \sin^2 x (1 - m^2)}$$

Now, put $\sin^2 x = t$

$$\Rightarrow 2 \sin x \cdot \cos x dx = dt$$

Here, at $x = 0 \Rightarrow t = 0$ and at $x = \frac{\pi}{2} \Rightarrow t = 1$

Therefore,

$$\Rightarrow I = \frac{1}{2} \int_0^1 \frac{dt}{1 - t(1 - m^2)}$$

$$\Rightarrow I = \frac{1}{2} \left[-\log|1-t(1-m^2)| \times \frac{1}{1-m^2} \right]_0^1$$

$$\Rightarrow I = -\frac{1}{2} \times \frac{1}{1-m^2} \left[\log|1-t(1-m^2)| \right]_0^1$$

$$\Rightarrow I = -\frac{1}{2(1-m^2)} \left[\log|1-1(1-m^2)| - \log|1-0(1-m^2)| \right]$$

$$\Rightarrow I = -\frac{1}{2(1-m^2)} \left[\log|1-1+m^2| - \log|1| \right]$$

$$\Rightarrow I = -\frac{1}{2(1-m^2)} \left[\log|m^2| \right]$$

$$\Rightarrow I = -\frac{1}{2(1-m^2)} 2 \times \log|m|$$

$$\Rightarrow I = -\frac{1}{(1-m^2)} \log|m|$$

Hence, $\int_0^{\frac{\pi}{2}} \frac{\tan x \, dx}{1+m^2 \tan^2 x} = -\frac{1}{(1-m^2)} \log|m|$.

30. $\int_1^2 \frac{dx}{\sqrt{(x-1)(2-x)}}$

Ans: Let $I = \int_1^2 \frac{dx}{\sqrt{(x-1)(2-x)}}$

$$\Rightarrow I = \int_1^2 \frac{dx}{\sqrt{2x-x^2-2+x}}$$

$$\Rightarrow I = \int_1^2 \frac{dx}{\sqrt{-x^2 + 3x - 2}}$$

$$\Rightarrow I = \int_1^2 \frac{dx}{\sqrt{-(x^2 - 3x + 2)}}$$

$$\Rightarrow I = \int_1^2 \frac{dx}{\sqrt{-\left[x^2 - 3x + \left(\frac{3}{2}\right)^2 - \left(\frac{3}{2}\right)^2 + 2\right]}}$$

$$\Rightarrow I = \int_1^2 \frac{dx}{\sqrt{-\left[\left\{x^2 - 3x + \left(\frac{3}{2}\right)^2\right\} - \left\{\frac{9}{4} - 2\right\}\right]}}$$

$$\Rightarrow I = \int_1^2 \frac{dx}{\sqrt{-\left[\left(x - \frac{3}{2}\right)^2 - \left(\frac{1}{2}\right)^2\right]}}$$

$$\Rightarrow I = \int_1^2 \frac{dx}{\sqrt{\left[\left(\frac{1}{2}\right)^2 - \left(x - \frac{3}{2}\right)^2\right]}}$$

$$\Rightarrow I = \left[\sin^{-1} \left(\frac{x - \frac{3}{2}}{\frac{1}{2}} \right) \right]_1^2$$

$$\Rightarrow I = \left[\sin^{-1} (2x - 3) \right]_1^2$$

$$\Rightarrow I = \left[\sin^{-1} (4 - 3) - \sin^{-1} (2 - 3) \right]$$

$$\Rightarrow I = \left[\sin^{-1} (1) - \sin^{-1} (-1) \right]$$

$$\Rightarrow I = \left[\frac{\pi}{2} - \left(-\frac{\pi}{2} \right) \right]$$

$$\Rightarrow I = \left[\frac{\pi}{2} + \frac{\pi}{2} \right]$$

$$\Rightarrow I = \pi$$

$$\text{Hence, } \int_1^2 \frac{dx}{\sqrt{(x-1)(2-x)}} = \pi$$

Evaluate the following:

$$31. \int_0^1 \frac{dx}{e^x + e^{-x}}$$

$$\text{Ans: Let } I = \int_0^1 \frac{dx}{e^x + e^{-x}}$$

$$\Rightarrow I = \int_0^1 \frac{dx}{e^x + \frac{1}{e^x}}$$

$$\Rightarrow I = \int_0^1 \frac{e^x \cdot dx}{(e^x)^2 + 1}$$

Now, put $e^x = t$

$$\Rightarrow e^x dx = dt$$

Now, at $x = 0 \Rightarrow t = 1$ and at $x = 1 \Rightarrow t = e$

Therefore,

$$\Rightarrow I = \int_1^e \frac{dt}{t^2 + 1}$$

$$\Rightarrow I = \left[\tan^{-1} t \right]_1^e$$

$$\Rightarrow I = (\tan^{-1} e - \tan^{-1} 1)$$

$$\Rightarrow I = \tan^{-1}e - \frac{\pi}{4}$$

Hence, $\int_0^1 \frac{dx}{e^x + e^{-x}} = \tan^{-1}e - \frac{\pi}{4}$

32. $\int_0^1 \frac{x dx}{\sqrt{1+x^2}}$

Ans: Let $I = \int_0^1 \frac{x dx}{\sqrt{1+x^2}}$

Now, put $1+x^2 = t^2$

$$\Rightarrow 2x dx = 2t dt$$

$$\Rightarrow x dx = t dt$$

here, at $x=0 \Rightarrow t=1$ and at $x=1 \Rightarrow t=\sqrt{2}$

Therefore,

$$\Rightarrow I = \int_1^{\sqrt{2}} \frac{t dt}{\sqrt{t^2}}$$

$$\Rightarrow I = \int_1^{\sqrt{2}} \frac{t dt}{t}$$

$$\Rightarrow I = \int_1^{\sqrt{2}} dt$$

$$\Rightarrow I = [t]_1^{\sqrt{2}}$$

$$\Rightarrow I = (\sqrt{2} - 1)$$

Hence, $\int_0^1 \frac{x dx}{\sqrt{1+x^2}} = (\sqrt{2} - 1)$

33. $\int_0^{\pi} x \sin x \cos^2 x \, dx$

Ans: Let $I = \int_0^{\pi} x \sin x \cos^2 x \, dx$ (i)

$\Rightarrow I = \int_0^{\pi} (\pi - x) \sin(\pi - x) \cos^2(\pi - x) \, dx$

$\Rightarrow I = \int_0^{\pi} (\pi - x) \sin x \cos^2 x \, dx$ (ii)

Adding eq(i) and (ii), we get

$\Rightarrow 2I = \pi \int_0^{\pi} \sin x \cos^2 x \, dx$

Now, put $\cos x = t$

$\Rightarrow -\sin x \, dx = dt$

$\Rightarrow \sin x \, dx = -dt$

here, at $x = 0 \Rightarrow t = 1$ and at $x = \pi \Rightarrow t = -1$

Therefore,

$\Rightarrow 2I = -\pi \int_1^{-1} t^2 \, dt$

$\Rightarrow 2I = -\pi \left[\frac{t^3}{3} \right]_1^{-1}$

$\Rightarrow 2I = -\frac{1}{3} \pi [t^3]_1^{-1}$

$\Rightarrow 2I = -\frac{1}{3} \pi (-1 - 1)$

$$\Rightarrow 2I = \frac{2\pi}{3}$$

$$\Rightarrow I = \frac{\pi}{3}$$

Hence, $\int_0^{\pi} x \sin x \cos^2 x \, dx = \frac{\pi}{3}$.

34. $\int_0^{\frac{1}{2}} \frac{dx}{(1+x^2)\sqrt{1-x^2}}$

Ans: Let $I = \int_0^{\frac{1}{2}} \frac{dx}{(1+x^2)\sqrt{1-x^2}}$

Put $x = \sin \theta$

$$\Rightarrow dx = \cos \theta \, d\theta$$

Here, at $x = 0 \Rightarrow \theta = 0$ and at $x = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{6}$

Therefore,

$$\Rightarrow I = \int_0^{\frac{\pi}{6}} \frac{\cos \theta \, d\theta}{(1+\sin^2 \theta)\sqrt{1-\sin^2 \theta}}$$

$$\Rightarrow I = \int_0^{\frac{\pi}{6}} \frac{\cos \theta \, d\theta}{(1+\sin^2 \theta)\cos \theta}$$

$$\Rightarrow I = \int_0^{\frac{\pi}{6}} \frac{d\theta}{(1+\sin^2 \theta)}$$

Divide by $\cos^2 \theta$ in numerator and denominator

$$\Rightarrow I = \int_0^{\frac{\pi}{6}} \frac{\frac{1}{\cos^2 \theta} d\theta}{\left(\frac{1}{\cos^2 \theta} + \frac{\sin^2 \theta}{\cos^2 \theta} \right)}$$

$$\Rightarrow I = \int_0^{\frac{\pi}{6}} \frac{\sec^2 \theta d\theta}{(\sec^2 \theta + \tan^2 \theta)}$$

$$\Rightarrow I = \int_0^{\frac{\pi}{6}} \frac{\sec^2 \theta d\theta}{(1 + \tan^2 \theta + \tan^2 \theta)}$$

$$\Rightarrow I = \int_0^{\frac{\pi}{6}} \frac{\sec^2 \theta d\theta}{(1 + 2\tan^2 \theta)}$$

Now, put $\tan \theta = t$

$$\Rightarrow \sec^2 \theta d\theta = dt$$

Here, at $\theta = 0 \Rightarrow t = 0$ and at $\theta = \frac{\pi}{6} \Rightarrow t = \frac{1}{\sqrt{3}}$

Therefore,

$$\Rightarrow I = \int_0^{\frac{1}{\sqrt{3}}} \frac{dt}{(1 + 2t^2)}$$

$$\Rightarrow I = \frac{1}{2} \int_0^{\frac{1}{\sqrt{3}}} \frac{dt}{\left(\frac{1}{2} + t^2 \right)}$$

$$\Rightarrow I = \frac{1}{2} \int_0^{\frac{1}{\sqrt{3}}} \frac{dt}{\left[\left(\frac{1}{\sqrt{2}} \right)^2 + t^2 \right]}$$

$$\Rightarrow I = \frac{1}{2} \left[\frac{1}{\frac{1}{\sqrt{2}}} \tan^{-1} \left(\frac{t}{\frac{1}{\sqrt{2}}} \right) \right]_0^{\frac{1}{\sqrt{3}}}$$

$$\Rightarrow I = \frac{1}{\sqrt{2}} \left[\tan^{-1} (\sqrt{2} \times t) \right]_0^{\frac{1}{\sqrt{3}}}$$

$$\Rightarrow I = \frac{1}{\sqrt{2}} \left[\tan^{-1} \left(\sqrt{2} \times \frac{1}{\sqrt{3}} \right) - \tan^{-1}(0) \right]$$

$$\Rightarrow I = \frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{\sqrt{2}}{\sqrt{3}} \right)$$

Long Answer (L.A)

35. $\int \frac{x^2 dx}{x^4 - x^2 - 12}$

Ans: Let $I = \int \frac{x^2 dx}{x^4 - x^2 - 12}$

$$\Rightarrow I = \int \frac{x^2 dx}{x^4 - 4x^2 + 3x^2 - 12}$$

$$\Rightarrow I = \int \frac{x^2 dx}{x^2(x^2 - 4) + 3(x^2 - 4)}$$

$$\Rightarrow I = \int \frac{x^2 dx}{(x^2 - 4)(x^2 + 3)}$$

$$\Rightarrow I = \int \frac{x^2 - 4 + 4 dx}{(x^2 - 4)(x^2 + 3)}$$

$$\Rightarrow I = \int \frac{(x^2 - 4) dx}{(x^2 - 4)(x^2 + 3)} + 4 \int \frac{dx}{(x^2 - 4)(x^2 + 3)}$$

$$\Rightarrow I = \int \frac{dx}{(x^2 + 3)} + 4 \int \frac{[(x^2 - 4) - (x^2 + 3)] dx}{(-7)(x^2 - 4)(x^2 + 3)}$$

$$\Rightarrow I = \int \frac{dx}{(x^2 + 3)} - \frac{4}{7} \int \frac{(x^2 - 4) dx}{(x^2 - 4)(x^2 + 3)} + \frac{4}{7} \int \frac{(x^2 + 3) dx}{(x^2 - 4)(x^2 + 3)}$$

$$\Rightarrow I = \int \frac{dx}{(x^2 + 3)} - \frac{4}{7} \int \frac{dx}{(x^2 + 3)} + \frac{4}{7} \int \frac{dx}{(x^2 - 4)}$$

$$\Rightarrow I = \frac{3}{7} \int \frac{dx}{(x^2 + 3)} + \frac{4}{7} \int \frac{dx}{(x^2 - 4)}$$

$$\Rightarrow I = \frac{3}{7} \int \frac{dx}{(x^2 + (\sqrt{3})^2)} + \frac{4}{7} \int \frac{dx}{(x^2 - 2^2)}$$

$$\Rightarrow I = \frac{3}{7} \left[\frac{1}{\sqrt{3}} \tan^{-1} \frac{x}{\sqrt{3}} \right] + \frac{4}{7} \cdot \frac{1}{2 \times 2} \log \left| \frac{x-2}{x+2} \right| + C$$

$$\Rightarrow I = \frac{\sqrt{3}}{7} \tan^{-1} \frac{x}{\sqrt{3}} + \frac{1}{7} \log \left| \frac{x-2}{x+2} \right| + C$$

Hence, $\int \frac{x^2 dx}{x^4 - x^2 - 12} = \frac{\sqrt{3}}{7} \tan^{-1} \frac{x}{\sqrt{3}} + \frac{1}{7} \log \left| \frac{x-2}{x+2} \right| + C.$

36. $\int \frac{x^2 dx}{(x^2 + a^2)(x^2 + b^2)}$

Ans: Let $I = \int \frac{x^2 + a^2 - a^2 dx}{(x^2 + a^2)(x^2 + b^2)}$

$$\Rightarrow I = \int \frac{(x^2 + a^2)dx}{(x^2 + a^2)(x^2 + b^2)} - \int \frac{a^2 dx}{(x^2 + a^2)(x^2 + b^2)}$$

$$\Rightarrow I = \int \frac{dx}{(x^2 + b^2)} - a^2 \int \frac{[(x^2 + a^2) - (x^2 + b^2)]dx}{(a^2 - b^2)(x^2 + a^2)(x^2 + b^2)}$$

$$\Rightarrow I = \int \frac{dx}{(x^2 + b^2)} - \frac{a^2}{(a^2 - b^2)} \int \frac{[(x^2 + a^2) - (x^2 + b^2)]dx}{(x^2 + a^2)(x^2 + b^2)}$$

$$\Rightarrow I = \int \frac{dx}{(x^2 + b^2)} - \frac{a^2}{(a^2 - b^2)} \left[\int \frac{(x^2 + a^2)dx}{(x^2 + a^2)(x^2 + b^2)} - \int \frac{(x^2 + b^2)dx}{(x^2 + a^2)(x^2 + b^2)} \right]$$

$$\Rightarrow I = \int \frac{dx}{(x^2 + b^2)} - \frac{a^2}{(a^2 - b^2)} \left[\int \frac{dx}{(x^2 + b^2)} - \int \frac{dx}{(x^2 + a^2)} \right]$$

$$\Rightarrow I = \frac{-b^2}{(a^2 - b^2)} \int \frac{dx}{(x^2 + b^2)} + \frac{a^2}{(a^2 - b^2)} \int \frac{dx}{(x^2 + a^2)}$$

$$\Rightarrow I = \frac{b^2}{(b^2 - a^2)} \frac{1}{b} \tan^{-1} \frac{x}{b} - \frac{a^2}{(b^2 - a^2)} \frac{1}{a} \tan^{-1} \frac{x}{a} + C$$

$$\Rightarrow I = \frac{b}{(b^2 - a^2)} \tan^{-1} \frac{x}{b} - \frac{a}{(b^2 - a^2)} \tan^{-1} \frac{x}{a} + C$$

37. $\int_0^{\pi} \frac{x}{1 + \sin x} dx$

Ans: Let $I = \int_0^{\pi} \frac{x}{1 + \sin x} dx$ (i)

$$\Rightarrow I = \int_0^{\pi} \frac{(\pi - x)}{1 + \sin(\pi - x)} dx$$

$$\Rightarrow I = \int_0^{\pi} \frac{(\pi - x)}{1 + \sin x} dx \dots\dots\dots(ii)$$

Now, adding eq(i) and (ii), we get

$$\Rightarrow 2I = \int_0^{\pi} \frac{\pi}{1 + \sin x} dx$$

$$\Rightarrow 2I = \pi \int_0^{\pi} \frac{(1 - \sin x)}{(1 + \sin x)(1 - \sin x)} dx$$

$$\Rightarrow 2I = \pi \int_0^{\pi} \frac{(1 - \sin x)}{(1 - \sin^2 x)} dx$$

$$\Rightarrow 2I = \pi \int_0^{\pi} \frac{(1 - \sin x)}{\cos^2 x} dx$$

$$\Rightarrow 2I = \pi \int_0^{\pi} \left(\frac{1}{\cos^2 x} - \frac{\sin x}{\cos^2 x} \right) dx$$

$$\Rightarrow 2I = \pi \int_0^{\pi} (\sec^2 x - \sec x \cdot \tan x) dx$$

$$\Rightarrow 2I = \pi [\tan x - \sec x]_0^{\pi}$$

$$\Rightarrow 2I = \pi [\tan \pi - \sec \pi - (\tan 0 - \sec 0)]$$

$$\Rightarrow 2I = \pi [0 - (-1) - (0 - 1)]$$

$$\Rightarrow 2I = \pi [1 + 1]$$

$$\Rightarrow 2I = 2\pi$$

$$\Rightarrow I = \pi$$

$$\text{Hence, } \int_0^{\pi} \frac{x}{1 + \sin x} dx = \pi$$

$$38. \int \frac{2x - 1}{(x - 1)(x + 2)(x - 3)} dx$$

Ans: Let $I = \int \frac{2x-1}{(x-1)(x+2)(x-3)} dx$

Now, $\frac{2x-1}{(x-1)(x+2)(x-3)} = \frac{A}{(x-1)} + \frac{B}{(x+2)} + \frac{C}{(x-3)}$

$\Rightarrow 2x-1 = A(x+2)(x-3) + B(x-1)(x-3) + C(x-1)(x+2)$

Here, put $x=1$ in above equation, we get

$\Rightarrow 1 = A(1+2)(1-3) + 0 + 0$

$\Rightarrow 1 = -6A$

$\Rightarrow A = -\frac{1}{6}$

Now, put $x=-2$

$\Rightarrow 2(-2)-1 = 0 + B(-2-1)(-2-3) + 0$

$\Rightarrow -5 = B(-3)(-5)$

$\Rightarrow B = -\frac{1}{3}$

Now, put $x=3$, we get

$\Rightarrow 2(3)-1 = 0 + 0 + C(3-1)(3+2)$

$\Rightarrow 5 = C(2)(5)$

$\Rightarrow C = \frac{1}{2}$

Therefore,

$\Rightarrow \int \frac{2x-1}{(x-1)(x+2)(x-3)} dx = -\int \frac{dx}{6(x-1)} - \int \frac{dx}{3(x+2)} + \int \frac{dx}{2(x-3)}$

$$= -\frac{1}{6} \log|(x-1)| - \frac{1}{3} \log|(x+2)| + \frac{1}{2} \log|(x-3)| + C$$

$$= -\log(x-1)^{\frac{1}{6}} - \log(x+2)^{\frac{1}{3}} + \log(x-3)^{\frac{1}{2}} + C$$

$$= \log \left| \frac{\sqrt{(x-3)}}{(x-1)^{\frac{1}{6}} (x+2)^{\frac{1}{3}}} \right| + C$$

Hence,
$$\int \frac{2x-1}{(x-1)(x+2)(x-3)} dx = \log \left| \frac{\sqrt{(x-3)}}{(x-1)^{\frac{1}{6}} (x+2)^{\frac{1}{3}}} \right| + C$$

39.
$$\int e^{\tan^{-1}x} \left(\frac{1+x+x^2}{1+x^2} \right) dx$$

Ans: Let
$$I = \int e^{\tan^{-1}x} \left(\frac{1+x+x^2}{1+x^2} \right) dx$$

$$\Rightarrow I = \int e^{\tan^{-1}x} \left(\frac{1+x^2}{1+x^2} + \frac{x}{1+x^2} \right) dx$$

$$\Rightarrow I = \int e^{\tan^{-1}x} \left(1 + \frac{x}{1+x^2} \right) dx$$

$$\Rightarrow I = \int e^{\tan^{-1}x} dx + \int \frac{xe^{\tan^{-1}x}}{1+x^2} dx$$

$$\Rightarrow I = I_1 + I_2 \dots\dots(i), \text{ where, } I_1 = \int e^{\tan^{-1}x} dx \text{ \& } I_2 = \int \frac{xe^{\tan^{-1}x}}{1+x^2} dx$$

Now,
$$I_2 = \int \frac{xe^{\tan^{-1}x}}{1+x^2} dx$$

Put $\tan^{-1}x = t$

$$\Rightarrow \frac{1}{1+x^2} dx = dt$$

Therefore,

$$\Rightarrow I_2 = \int e^t \cdot \tan t \, dt$$

$$\Rightarrow I_2 = \int e^t \cdot \tan t \, dt$$

$$\Rightarrow I_2 = \tan t \int e^t \, dt - \int \left(\frac{d}{dx} \tan t \cdot \int e^t \, dt \right) dt$$

$$\Rightarrow I_2 = \tan t \cdot e^t - \int \sec^2 t \cdot e^t \, dt + C$$

$$\Rightarrow I_2 = x \cdot e^{\tan^{-1}x} - \int (1 + \tan^2 t) \cdot e^{\tan^{-1}x} \frac{1}{1+x^2} dx + C$$

$$\Rightarrow I_2 = x \cdot e^{\tan^{-1}x} - \int (1 + x^2) \cdot e^{\tan^{-1}x} \frac{1}{1+x^2} dx + C$$

$$\Rightarrow I_2 = x \cdot e^{\tan^{-1}x} - \int e^{\tan^{-1}x} dx + C$$

Put values of I_1 and I_2 , in equation (i),

$$\Rightarrow I = \int e^{\tan^{-1}x} dx + x \cdot e^{\tan^{-1}x} - \int e^{\tan^{-1}x} dx + C$$

$$\Rightarrow I = x \cdot e^{\tan^{-1}x} + C$$

$$\text{Hence, value of } \int e^{\tan^{-1}x} \left(\frac{1+x+x^2}{1+x^2} \right) dx = x \cdot e^{\tan^{-1}x} + C.$$

40. $\int \sin^{-1} \sqrt{\frac{x}{a+x}} dx$

Ans: Let $I = \int \sin^{-1} \sqrt{\frac{x}{a+x}} dx$

Put $x = a \tan^2 \theta$

$$\Rightarrow dx = a \cdot 2 \tan \theta \cdot \sec^2 \theta d\theta$$

Therefore,

$$\Rightarrow I = \int \sin^{-1} \sqrt{\frac{a \tan^2 \theta}{a + a \tan^2 \theta}} \times 2a \tan \theta \cdot \sec^2 \theta d\theta$$

$$\Rightarrow I = \int \sin^{-1} \sqrt{\frac{a \tan^2 \theta}{a(1 + \tan^2 \theta)}} \times 2a \tan \theta \cdot \sec^2 \theta d\theta$$

$$\Rightarrow I = \int \sin^{-1} \sqrt{\frac{\tan^2 \theta}{(1 + \tan^2 \theta)}} \times 2a \tan \theta \cdot \sec^2 \theta d\theta$$

$$\Rightarrow I = \int \sin^{-1} \sqrt{\frac{\tan^2 \theta}{\sec^2 \theta}} \times 2a \tan \theta \cdot \sec^2 \theta d\theta$$

$$\Rightarrow I = \int \sin^{-1} \sqrt{\sin^2 \theta} \times 2a \tan \theta \cdot \sec^2 \theta d\theta$$

$$\Rightarrow I = \int \sin^{-1}(\sin \theta) \times 2a \tan \theta \cdot \sec^2 \theta d\theta$$

$$\Rightarrow I = 2a \int \theta \tan \theta \cdot \sec^2 \theta d\theta$$

$$\Rightarrow I = 2a \left[\theta \int \tan \theta \cdot \sec^2 \theta d\theta - \int \left(\frac{d}{d\theta} \theta \cdot \int \tan \theta \cdot \sec^2 \theta d\theta \right) d\theta \right]$$

Put $\tan \theta = t$

$$\Rightarrow \sec^2 \theta d\theta = dt$$

$$\Rightarrow I = 2a \left[\theta \int t dt - \int \left(\int t dt \right) d\theta \right]$$

$$\Rightarrow I = 2a \left[\theta \cdot \frac{t^2}{2} - \int \frac{t^2}{2} d\theta \right]$$

$$\Rightarrow I = 2a \left[\theta \cdot \frac{\tan^2 \theta}{2} - \int \frac{\tan^2 \theta}{2} d\theta \right]$$

$$\Rightarrow I = a \left[\theta (\tan^2 \theta) - \int \tan^2 \theta d\theta \right]$$

$$\Rightarrow I = a\theta(\tan^2\theta) - a \int (\sec^2\theta - 1)d\theta$$

$$\Rightarrow I = a\theta(\tan^2\theta) - a \tan\theta + a\theta + C$$

$$\Rightarrow I = a \left[\frac{x}{a} \left(\tan^{-1} \sqrt{\frac{x}{a}} \right) - \sqrt{\frac{x}{a}} + \tan^{-1} \sqrt{\frac{x}{a}} \right] + C$$

Hence, $\int \sin^{-1} \sqrt{\frac{x}{a+x}} dx = a \left[\frac{x}{a} \left(\tan^{-1} \sqrt{\frac{x}{a}} \right) - \sqrt{\frac{x}{a}} + \tan^{-1} \sqrt{\frac{x}{a}} \right] + C$.

41. $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{\sqrt{1+\cos x}}{(1-\cos x)^{\frac{5}{2}}} dx \dots$

Ans: Let $I = \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{\sqrt{1+\cos x}}{(1-\cos x)^{\frac{5}{2}}} dx$

$$I = \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{\sqrt{2\cos^2 \frac{x}{2}}}{\left(2\sin^2 \frac{x}{2} \right)^{\frac{5}{2}}} dx$$

$$\Rightarrow I = \frac{\sqrt{2}}{2^{\frac{5}{2}}} \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{\cos \frac{x}{2}}{\sin^5 \frac{x}{2}} dx$$

Put $\sin \frac{x}{2} = t$

$$\Rightarrow \frac{1}{2} \cos \frac{x}{2} dx = dt$$

$$\Rightarrow \cos \frac{x}{2} dx = 2dt$$

Here, at $x = \frac{\pi}{3} \Rightarrow t = \frac{1}{2}$ and at $x = \frac{\pi}{2} \Rightarrow t = \frac{1}{\sqrt{2}}$

Therefore,

$$\Rightarrow I = \frac{\sqrt{2}}{\sqrt{32}} \int_{\frac{1}{2}}^{\frac{1}{\sqrt{2}}} \frac{2 dt}{t^5}$$

$$\Rightarrow I = \frac{\sqrt{2}}{4\sqrt{2}} \times 2 \int_{\frac{1}{2}}^{\frac{1}{\sqrt{2}}} t^{-5} dt$$

$$\Rightarrow I = \frac{1}{2} \left[\frac{t^{-4}}{-4} \right]_{\frac{1}{2}}^{\frac{1}{\sqrt{2}}}$$

$$\Rightarrow I = -\frac{1}{8} \left[t^{-4} \right]_{\frac{1}{2}}^{\frac{1}{\sqrt{2}}}$$

$$\Rightarrow I = -\frac{1}{8} \left[\left(\frac{1}{\sqrt{2}} \right)^{-4} - \left(\frac{1}{2} \right)^{-4} \right]$$

$$\Rightarrow I = -\frac{1}{8} \left[(\sqrt{2})^4 - (2)^4 \right]$$

$$\Rightarrow I = -\frac{1}{8} [4 - 16]$$

$$\Rightarrow I = \frac{1}{8} \times 12$$

$$\Rightarrow I = \frac{3}{2}$$

$$\text{Hence, } \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{\sqrt{1 + \cos x}}{(1 - \cos x)^{\frac{5}{2}}} dx = \frac{3}{2}.$$

42. $\int e^{-3x} \cos^3 x \, dx$

Ans: Let $I = \int e^{-3x} \cos^3 x \, dx$

$$\Rightarrow I = \int e^{-3x} \left(\frac{\cos 3x + 3 \cos x}{4} \right) dx$$

$$\Rightarrow I = \frac{1}{4} \int (e^{-3x} \cos 3x + 3e^{-3x} \cos x) dx$$

$$\Rightarrow I = \frac{1}{4} \int e^{-3x} \cos 3x \, dx + \frac{3}{4} \int e^{-3x} \cos x \, dx$$

$$\Rightarrow I = \frac{1}{4} I_1 + \frac{3}{4} I_2 \dots\dots\dots \text{eq (i)}$$

Where, $I_1 = \int e^{-3x} \cos 3x \, dx$ and $I_2 = \int 3e^{-3x} \cos x \, dx$

Now, $I_1 = \int e^{-3x} \cos 3x \, dx$

$$\Rightarrow I_1 = \cos 3x \int e^{-3x} \, dx - \int \left(\frac{d}{dx} \cos 3x \cdot \int e^{-3x} \, dx \right) dx$$

$$\Rightarrow I_1 = \frac{e^{-3x}}{-3} \cos 3x - \int \left(-3 \sin 3x \cdot \frac{e^{-3x}}{-3} \right) dx$$

$$\Rightarrow I_1 = \frac{e^{-3x}}{-3} \cos 3x - \int (e^{-3x} \sin 3x) dx$$

$$\Rightarrow I_1 = \frac{e^{-3x}}{-3} \cos 3x - \left[\sin 3x \int e^{-3x} \, dx - \int \left(\frac{d}{dx} \sin 3x \cdot \int e^{-3x} \, dx \right) dx \right]$$

$$\Rightarrow I_1 = \frac{e^{-3x}}{-3} \cos 3x - \left[\frac{e^{-3x}}{-3} \sin 3x - \int \left(3 \cos 3x \cdot \frac{e^{-3x}}{-3} \right) dx \right]$$

$$\Rightarrow I_1 = \frac{e^{-3x}}{-3} \cos 3x - \left[\frac{e^{-3x}}{-3} \sin 3x + \int (e^{-3x} \cos 3x) dx \right]$$

$$\Rightarrow I_1 = \frac{e^{-3x}}{-3} \cos 3x - \left[\frac{e^{-3x}}{-3} \sin 3x + I_1 \right]$$

$$\Rightarrow I_1 = \frac{e^{-3x}}{-3} \cos 3x + \frac{e^{-3x}}{3} \sin 3x - I_1$$

$$\Rightarrow 2I_1 = \frac{e^{-3x}}{-3} \cos 3x + \frac{e^{-3x}}{3} \sin 3x$$

$$\Rightarrow 2I_1 = \frac{1}{3} e^{-3x} (-\cos 3x + \sin 3x) + C_1$$

$$\Rightarrow I_1 = \frac{1}{6} e^{-3x} (\sin 3x - \cos 3x) + C_1$$

Now, $I_2 = \int e^{-3x} \cos x dx$

$$\Rightarrow I_2 = \int e^{-3x} \cos x dx$$

$$\Rightarrow I_2 = \cos x \int e^{-3x} dx - \int \left(\frac{d}{dx} \cos x \cdot \int e^{-3x} dx \right) dx$$

$$\Rightarrow I_2 = \frac{e^{-3x}}{-3} \cos x - \int \left(-\sin x \cdot \frac{e^{-3x}}{-3} \right) dx$$

$$\Rightarrow I_2 = \frac{e^{-3x}}{-3} \cos x - \frac{1}{3} \int (e^{-3x} \cdot \sin x) dx$$

$$\Rightarrow I_2 = \frac{e^{-3x}}{-3} \cos x - \frac{1}{3} \left[\sin x \int e^{-3x} dx - \int \left(\frac{d}{dx} \sin x \cdot \int e^{-3x} dx \right) dx \right]$$

$$\Rightarrow I_2 = \frac{e^{-3x}}{-3} \cos x - \frac{1}{3} \left[\frac{e^{-3x}}{-3} \sin x - \int \left(\frac{e^{-3x}}{-3} \cos x \right) dx \right]$$

$$\Rightarrow I_2 = \frac{e^{-3x}}{-3} \cos x - \frac{1}{3} \left[\frac{e^{-3x}}{-3} \sin x + \frac{1}{3} \int (e^{-3x} \cos x) dx \right]$$

$$\Rightarrow I_2 = \frac{e^{-3x}}{-3} \cos x + \frac{1}{9} e^{-3x} \sin x - \frac{1}{9} \int (e^{-3x} \cos x) dx$$

$$\Rightarrow I_2 = \frac{e^{-3x}}{-3} \cos x + \frac{1}{9} e^{-3x} \sin x - \frac{1}{9} I_2$$

$$\Rightarrow I_2 + \frac{I_2}{9} = \frac{e^{-3x}}{-3} \cos x + \frac{1}{9} e^{-3x} \sin x$$

$$\Rightarrow \frac{10I_2}{9} = \frac{-3e^{-3x} \cos x + e^{-3x} \sin x}{9} + C_2$$

$$\Rightarrow 10I_2 = e^{-3x} (\sin x - 3 \cos x) + C_2$$

$$\Rightarrow I_2 = \frac{1}{10} e^{-3x} (\sin x - 3 \cos x) + C_2$$

Now, put the value of I_1 and I_2 in eq(i), we get

$$\Rightarrow I = \frac{1}{24} e^{-3x} (\sin 3x - \cos 3x) + \frac{3}{40} e^{-3x} (\sin x - 3 \cos x) + C$$

43. $\int \sqrt{\tan x} dx$

Ans: Let $I = \int \sqrt{\tan x} dx$

Put $\tan x = t^2$

$$\Rightarrow \sec^2 x dx = 2t dt$$

$$\Rightarrow (1 + \tan^2 x) dx = 2t dt$$

$$\Rightarrow dx = \frac{2t}{(1 + \tan^2 x)} dt$$

$$\Rightarrow dx = \frac{2t}{(1+t^4)} dt$$

Therefore,

$$\Rightarrow I = \int \sqrt{t^2} \cdot \frac{2t}{(1+t^4)} dt$$

$$\Rightarrow I = \int \frac{2t^2}{(1+t^4)} dt$$

$$\Rightarrow I = \int \frac{(1+t^2) - (1-t^2)}{(1+t^4)} dt$$

$$\Rightarrow I = \int \frac{(1+t^2)}{(1+t^4)} dt - \int \frac{(1-t^2)}{(1+t^4)} dt$$

$$\Rightarrow I = \int \frac{\left(\frac{1}{t^2} + 1\right)}{\left(\frac{1}{t^2} + t^2\right)} dt - \int \frac{\left(\frac{1}{t^2} - 1\right)}{\left(\frac{1}{t^2} + t^2\right)} dt$$

$$\Rightarrow I = \int \frac{\left(\frac{1}{t^2} + 1\right)}{\left(t - \frac{1}{t}\right)^2 + 2} dt - \int \frac{\left(\frac{1}{t^2} - 1\right)}{\left(t + \frac{1}{t}\right)^2 - 2} dt$$

$$\text{Put } u = \left(t - \frac{1}{t}\right) \Rightarrow du = \left(\frac{1}{t^2} + 1\right) dt$$

$$\text{And } v = \left(t + \frac{1}{t}\right) \Rightarrow dv = \left(\frac{1}{t^2} - 1\right) dt$$

Therefore,

$$\Rightarrow I = \int \frac{du}{(u)^2 + 2} - \int \frac{-dv}{(v)^2 - 2}$$

$$\Rightarrow I = \int \frac{du}{(u)^2 + (\sqrt{2})^2} + \int \frac{dv}{(v)^2 - (\sqrt{2})^2}$$

$$\Rightarrow I = \frac{1}{\sqrt{2}} \tan^{-1} \frac{u}{\sqrt{2}} + \frac{1}{2\sqrt{2}} \log \left| \frac{v - \sqrt{2}}{v + \sqrt{2}} \right| + C$$

$$\Rightarrow I = \frac{1}{\sqrt{2}} \tan^{-1} \frac{\left(t - \frac{1}{t}\right)}{\sqrt{2}} + \frac{1}{2\sqrt{2}} \log \left| \frac{\left(t + \frac{1}{t}\right) - \sqrt{2}}{\left(t + \frac{1}{t}\right) + \sqrt{2}} \right| + C$$

$$\Rightarrow I = \frac{1}{\sqrt{2}} \tan^{-1} \frac{\left(\sqrt{\tan x} - \frac{1}{\sqrt{\tan x}}\right)}{\sqrt{2}} + \frac{1}{2\sqrt{2}} \log \left| \frac{\left(\sqrt{\tan x} + \frac{1}{\sqrt{\tan x}}\right) - \sqrt{2}}{\left(\sqrt{\tan x} + \frac{1}{\sqrt{\tan x}}\right) + \sqrt{2}} \right| + C$$

$$\Rightarrow I = \frac{1}{\sqrt{2}} \tan^{-1} \frac{(\tan x - 1)}{\sqrt{2 \tan x}} + \frac{1}{2\sqrt{2}} \log \left| \frac{\tan x - \sqrt{2 \tan x} + 1}{\tan x + \sqrt{2 \tan x} + 1} \right| + C$$

44. $\int_0^{\frac{\pi}{2}} \frac{dx}{(a^2 \cos^2 x + b^2 \sin^2 x)^2}$

Ans: Let $I = \int_0^{\frac{\pi}{2}} \frac{dx}{(a^2 \cos^2 x + b^2 \sin^2 x)^2}$

Divide numerator and denominator by $\cos^4 x$

$$\Rightarrow I = \int_0^{\frac{\pi}{2}} \frac{\frac{dx}{\cos^4 x}}{(a^2 \cos^2 x + b^2 \sin^2 x)^2}$$

$$\Rightarrow \frac{dx}{\cos^4 x}$$

$$\Rightarrow I = \int_0^{\frac{\pi}{2}} \frac{\sec^4 x \, dx}{(a^2 + b^2 \tan^2 x)^2}$$

$$\Rightarrow I = \int_0^{\frac{\pi}{2}} \frac{(1 + \tan^2 x) \sec^2 x \, dx}{(a^2 + b^2 \tan^2 x)^2}$$

Put $b \tan x = a \tan t$

$$\Rightarrow b \sec^2 x \, dx = a \sec^2 t \, dt$$

$$\Rightarrow \sec^2 x \, dx = \frac{a}{b} \sec^2 t \, dt$$

When $x \rightarrow 0, t \rightarrow 0$ and When $x \rightarrow \frac{\pi}{2}, \tan t \rightarrow 0, t \rightarrow \frac{\pi}{2}$

Therefore,

$$\Rightarrow I = \int_0^{\frac{\pi}{2}} \frac{\left(1 + \left(\frac{a}{b} \tan t\right)^2\right)}{a^4 \sec^4 t} \frac{a}{b} \sec^2 t \, dt$$

$$\Rightarrow I = \frac{1}{a^3 b^3} \int_0^{\frac{\pi}{2}} (b^2 + a^2 \tan^2 t) \cos^2 t \, dt$$

$$\Rightarrow I = \frac{1}{a^3 b^3} \int_0^{\frac{\pi}{2}} (b^2 \cos^2 t + a^2 \sin^2 t) \, dt$$

$$\Rightarrow I = \frac{1}{a^3 b^3} \int_0^{\frac{\pi}{2}} \left(b^2 \frac{1 + \cos 2t}{2} + a^2 \frac{1 - \cos 2t}{2} \right) dt$$

$$\Rightarrow I = \frac{1}{2a^3 b^3} \int_0^{\frac{\pi}{2}} [(a^2 + b^2) + (b^2 - a^2) \cos 2t] \, dt$$

$$\Rightarrow I = \frac{1}{2a^3 b^3} \left[(a^2 + b^2)x - \frac{1}{2}(b^2 - a^2) \sin 2t \right]_0^{\frac{\pi}{2}}$$

$$\Rightarrow I = \frac{1}{2a^3b^3} \left[(a^2 + b^2) \frac{\pi}{2} - \frac{1}{2} (b^2 - a^2) \sin \pi \right]$$

$$\Rightarrow I = \frac{1}{2a^3b^3} \left[(a^2 + b^2) \frac{\pi}{2} \right]$$

$$\Rightarrow I = \frac{\pi(a^2 + b^2)}{4a^3b^3}$$

Hence, $\int_0^{\frac{\pi}{2}} \frac{dx}{(a^2 \cos^2 x + b^2 \sin^2 x)^2} = \frac{\pi(a^2 + b^2)}{4a^3b^3}.$

45. $\int_0^1 x \log(1+2x) dx$

Ans: Let $I = \int_0^1 x \log(1+2x) dx$

$$\Rightarrow I = \left[\log(1+2x) \int_0^1 x dx \right]_0^1 - \int_0^1 \left(\frac{d}{dx} \log(1+2x) \cdot \int_0^1 x dx \right) dx$$

$$\Rightarrow I = \left[\frac{x^2}{2} \log(1+2x) \right]_0^1 - \int_0^1 \left(\frac{2}{1+2x} \cdot \frac{x^2}{2} \right) dx$$

$$\Rightarrow I = \left(\frac{1}{2} \log 3 - 0 \right) - \int_0^1 \left(\frac{x^2}{1+2x} \right) dx$$

$$\Rightarrow I = \left(\frac{1}{2} \log 3 - 0 \right) - \int_0^1 \left(\frac{\frac{x}{2}}{\frac{x}{2} + 1} \right) dx$$

$$\Rightarrow I = \left(\frac{1}{2} \log 3 \right) - \frac{1}{2} \int_0^1 x dx + \frac{1}{2} \int_0^1 \frac{x}{1+2x} dx$$

$$\Rightarrow I = \frac{1}{2} \log 3 - \frac{1}{2} \left(\frac{1}{2} - 0 \right) + \frac{1}{4} \int_0^1 \frac{1+2x-1}{1+2x} dx$$

$$\Rightarrow I = \frac{1}{2} \log 3 - \frac{1}{4} + \frac{1}{4} \int_0^1 \frac{1+2x}{1+2x} dx - \frac{1}{4} \int_0^1 \frac{dx}{1+2x}$$

$$\Rightarrow I = \frac{1}{2} \log 3 - \frac{1}{4} + \frac{1}{4} \int_0^1 dx - \frac{1}{4} \left[\frac{1}{2} \log(1+2x) \right]_0^1$$

$$\Rightarrow I = \frac{1}{2} \log 3 - \frac{1}{4} + \frac{1}{4} (x)_0^1 - \frac{1}{8} [\log(3) - 0]$$

$$\Rightarrow I = \frac{1}{2} \log 3 - \frac{1}{4} + \frac{1}{4} - \frac{1}{8} [\log(3)]$$

$$\Rightarrow I = \frac{1}{2} \log 3 - \frac{1}{8} \log 3$$

$$\Rightarrow I = \frac{3}{8} \log 3$$

Hence, $\int_0^1 x \log(1+2x) dx = \frac{3}{8} \log 3.$

46. $\int_0^\pi x \log \sin x \, dx$

Ans: Let $I = \int_0^\pi x \log \sin x \, dx$ (i)

$$\Rightarrow I = \int_0^\pi (\pi - x) \log \sin(\pi - x) \, dx$$

$$\Rightarrow I = \int_0^\pi (\pi - x) \log \sin x \, dx$$
 (ii)

Adding eq (i) and (ii), we get

$$\Rightarrow 2I = \pi \int_0^{\pi} \log \sin x \, dx$$

$$\Rightarrow 2I = 2\pi \int_0^{\frac{\pi}{2}} \log \sin x \, dx$$

$$\Rightarrow I = \pi \int_0^{\frac{\pi}{2}} \log \sin x \, dx \dots\dots\dots(iii)$$

$$\Rightarrow I = \pi \int_0^{\frac{\pi}{2}} \log \sin \left(\frac{\pi}{2} - x \right) dx$$

$$\Rightarrow I = \pi \int_0^{\frac{\pi}{2}} \log \cos x \, dx \dots\dots\dots(iv)$$

Adding eq (iii) and (iv), we get

$$\Rightarrow 2I = \pi \int_0^{\frac{\pi}{2}} (\log \sin x + \log \cos x) dx$$

$$\Rightarrow 2I = \pi \int_0^{\frac{\pi}{2}} (\log \sin x \cdot \cos x) dx$$

$$\Rightarrow 2I = \pi \int_0^{\frac{\pi}{2}} \left(\log \frac{2}{2} \sin x \cdot \cos x \right) dx$$

$$\Rightarrow 2I = \pi \int_0^{\frac{\pi}{2}} \left(\log \frac{\sin 2x}{2} \right) dx$$

$$\Rightarrow 2I = \pi \int_0^{\frac{\pi}{2}} (\log \sin 2x - \log 2) dx$$

$$\Rightarrow 2I = \pi \int_0^{\frac{\pi}{2}} \log \sin 2x \, dx - \pi \int_0^{\frac{\pi}{2}} \log 2 \, dx$$

$$\Rightarrow 2I = \pi \int_0^{\frac{\pi}{2}} \log \sin 2x \, dx - \pi \log 2 \times \frac{\pi}{2}$$

Put $2x = t \Rightarrow dx = \frac{dt}{2}$

Here, at $x = 0 \rightarrow t = 0$ and at $x = \frac{\pi}{2} \rightarrow t = \pi$

$$\Rightarrow 2I = \frac{\pi}{2} \int_0^{\pi} \log \sin t \, dt - \frac{\pi^2}{2} \log 2$$

$$\Rightarrow 2I = \frac{\pi}{2} \times 2 \int_0^{\frac{\pi}{2}} \log \sin t \, dt - \frac{\pi^2}{2} \log 2$$

$$\Rightarrow 2I = \pi \int_0^{\frac{\pi}{2}} \log \sin t \, dt - \frac{\pi^2}{2} \log 2$$

$$\Rightarrow 2I = I - \frac{\pi^2}{2} \log 2 \quad [\text{from eq (iii)}]$$

$$\Rightarrow I = -\frac{\pi^2}{2} \log 2$$

Hence, $\int_0^{\pi} x \log \sin x \, dx = -\frac{\pi^2}{2} \log 2.$

47. $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \log(\sin x + \cos x) \, dx$

$$I = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \log(\sin x + \cos x) \, dx$$

Ans: Let $\frac{\pi}{4}$ (i)

$$\Rightarrow I = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \log \left[\sin \left(\frac{\pi}{4} - \frac{\pi}{4} - x \right) + \cos \left(\frac{\pi}{4} - \frac{\pi}{4} - x \right) \right] dx$$

$$\Rightarrow I = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \log [\sin(-x) + \cos(-x)] dx$$

$$\Rightarrow I = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \log (-\sin x + \cos x) dx$$

$$\Rightarrow I = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \log (\cos x - \sin x) dx \dots\dots\dots(ii)$$

Adding eq (i) and (ii), we get

$$\Rightarrow 2 I = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \log (\sin x + \cos x) dx + \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \log (\cos x - \sin x) dx$$

$$\Rightarrow 2 I = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \log [(\sin x + \cos x) \cdot (\cos x - \sin x)] dx$$

$$\Rightarrow 2 I = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \log [\cos^2 x - \sin^2 x] dx$$

$$\Rightarrow 2 I = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \log \cos 2x dx$$

$$\Rightarrow 2 I = 2 \int_0^{\frac{\pi}{4}} \log \cos 2x dx$$

$$\Rightarrow I = \int_0^{\frac{\pi}{4}} \log \cos 2x dx \dots\dots\dots (iii)$$

Now, put $2x = t \Rightarrow dx = \frac{dt}{2}$

Here, at $x = 0 \rightarrow t = 0$ and at $x = \frac{\pi}{4} \rightarrow t = \frac{\pi}{2}$

Therefore,

$$\Rightarrow I = \frac{1}{2} \int_0^{\frac{\pi}{2}} \log \cos t dt \dots\dots\dots (iv)$$

$$\Rightarrow I = \frac{1}{2} \int_0^{\frac{\pi}{2}} \log \cos \left(\frac{\pi}{2} - t \right) dt$$

$$\Rightarrow I = \frac{1}{2} \int_0^{\frac{\pi}{2}} \log \sin t dt \dots\dots\dots (v)$$

Adding eq (iv) and (v), we get

$$\Rightarrow 2I = \frac{1}{2} \int_0^{\frac{\pi}{2}} \log \cos t dt + \frac{1}{2} \int_0^{\frac{\pi}{2}} \log \sin t dt$$

$$\Rightarrow 2I = \frac{1}{2} \int_0^{\frac{\pi}{2}} \log \sin t \cos t dt$$

$$\Rightarrow 4I = \int_0^{\frac{\pi}{2}} \log \frac{2}{2} \sin t \cos t dt$$

$$\Rightarrow 4I = \int_0^{\frac{\pi}{2}} \log \frac{\sin 2t}{2} dt$$

$$\Rightarrow 4I = \int_0^{\frac{\pi}{2}} (\log \sin 2t - \log 2) dt$$

$$\Rightarrow 4I = \int_0^{\frac{\pi}{2}} \left(\log \sin \left(\frac{\pi}{2} - 2t \right) \right) dt - \int_0^{\frac{\pi}{2}} \log 2 dt$$

$$\Rightarrow 4I = \int_0^{\frac{\pi}{2}} (\log \cos 2t) dt - \frac{\pi}{2} \log 2$$

$$\Rightarrow 4I = 2 \int_0^{\frac{\pi}{4}} (\log \cos 2t) dt - \frac{\pi}{2} \log 2$$

$$\Rightarrow 4I = 2I - \frac{\pi}{2} \log 2$$

$$\Rightarrow 2I = -\frac{\pi}{2} \log 2$$

$$\Rightarrow I = -\frac{\pi}{4} \log 2$$

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \log(\sin x + \cos x) dx = -\frac{\pi}{4} \log 2.$$

Hence,

Objective Type Questions

Choose the correct option from given four options in each of the Exercises from 48 to 63.

48. $\int \frac{\cos 2x - \cos 2\theta}{\cos x - \cos \theta} dx$ is equal to

- (A) $2(\sin x + x \cos \theta) + C$
- (B) $2(\sin x - x \cos \theta) + C$
- (C) $2(\sin x + 2x \cos \theta) + C$
- (D) $2(\sin x - 2x \cos \theta) + C$

$$\begin{aligned} \text{Ans: Let } I &= \int \frac{\cos 2x - \cos 2\theta}{\cos x - \cos \theta} dx \\ \Rightarrow I &= \int \frac{2 \cos^2 x - 1 - (2 \cos^2 \theta - 1)}{\cos x - \cos \theta} dx \\ \Rightarrow I &= \int \frac{2 \cos^2 x - 1 - 2 \cos^2 \theta + 1}{\cos x - \cos \theta} dx \\ \Rightarrow I &= \int \frac{2 \cos^2 x - 2 \cos^2 \theta}{\cos x - \cos \theta} dx \\ \Rightarrow I &= \int \frac{2(\cos^2 x - \cos^2 \theta)}{(\cos x - \cos \theta)} dx \\ \Rightarrow I &= \int \frac{2(\cos^2 x - \cos^2 \theta)}{(\cos x - \cos \theta)} dx \\ \Rightarrow I &= 2 \int (\cos x + \cos \theta) dx \\ \Rightarrow I &= 2(\sin x + x \cos \theta) + C \end{aligned}$$

Hence, option (A) is correct answer.

49. $\int \frac{dx}{\sin(x-a)\sin(x-b)}$ is equal to

(A) $\sin(b-a) \log \left| \frac{\sin(x-b)}{\sin(x-a)} \right| + C$

(B) $\operatorname{cosec}(b-a) \log \left| \frac{\sin(x-a)}{\sin(x-b)} \right| + C$

(C) $\operatorname{cosec}(b-a) \log \left| \frac{\sin(x-b)}{\sin(x-a)} \right| + C$

$$(D) \quad \sin(b-a) \log \left| \frac{\sin(x-b)}{\sin(x-a)} \right| + C$$

Ans: Let $I = \int \frac{dx}{\sin(x-a)\sin(x-b)}$

$$\Rightarrow I = \frac{1}{\sin(b-a)} \int \frac{\sin(b-a) dx}{\sin(x-a)\sin(x-b)}$$

$$\Rightarrow I = \frac{1}{\sin(b-a)} \int \frac{\sin(x-a-x+b)}{\sin(x-a)\sin(x-b)} dx$$

$$\Rightarrow I = \frac{1}{\sin(b-a)} \int \frac{\sin\{(x-a)-(x-b)\}}{\sin(x-a)\sin(x-b)} dx$$

$$\Rightarrow I = \frac{1}{\sin(b-a)} \int \frac{\sin(x-a) \cdot \cos(x-b) - \sin(x-b) \cdot \cos(x-a)}{\sin(x-a)\sin(x-b)} dx$$

$$\Rightarrow I = \frac{1}{\sin(b-a)} \int \frac{\cos(x-b)}{\sin(x-b)} dx - \frac{1}{\sin(b-a)} \int \frac{\cos(x-a)}{\sin(x-a)} dx$$

$$\Rightarrow I = \frac{1}{\sin(b-a)} \int \cot(x-b) dx - \frac{1}{\sin(b-a)} \int \cot(x-a) dx$$

$$\Rightarrow I = \frac{1}{\sin(b-a)} \log|\sin(x-b)| - \frac{1}{\sin(b-a)} \log|\sin(x-a)| + C$$

$$\Rightarrow I = \frac{1}{\sin(b-a)} \left[\log|\sin(x-b)| - \log|\sin(x-a)| \right] + C$$

$$\Rightarrow I = \frac{1}{\sin(b-a)} \log \left| \frac{(x-b)}{(x-a)} \right| + C$$

$$\Rightarrow I = \operatorname{cosec}(b-a) \log \left| \frac{(x-b)}{(x-a)} \right| + C$$

Hence, option (C) is correct answer.

50. $\int \tan^{-1} \sqrt{x} \, dx$ is equal to

(A) $(x+1)\tan^{-1} \sqrt{x} - \sqrt{x} + C$

(B) $x\tan^{-1} \sqrt{x} - \sqrt{x} + C$

(C) $\sqrt{x} - x\tan^{-1} \sqrt{x} + C$

(D) $\sqrt{x} - (x+1)\tan^{-1} \sqrt{x} + C$

Ans: Let $I = \int \tan^{-1} \sqrt{x} \, dx$

$$\Rightarrow I = \int 1 \cdot \tan^{-1} \sqrt{x} \, dx$$

$$\Rightarrow I = \tan^{-1} \sqrt{x} \int 1 \, dx - \int \left(\frac{d}{dx} \tan^{-1} \sqrt{x} \cdot \int 1 \, dx \right) dx$$

$$\Rightarrow I = x \tan^{-1} \sqrt{x} - \int \left(\frac{1}{1+x} \cdot \frac{1}{2\sqrt{x}} \cdot x \right) dx$$

$$\Rightarrow I = x \tan^{-1} \sqrt{x} - \frac{1}{2} \int \left(\frac{\sqrt{x}}{1+x} \right) dx$$

Now, put $x = t^2$

$$\Rightarrow dx = 2t \, dt$$

Therefore,

$$\Rightarrow I = x \tan^{-1} \sqrt{x} - \frac{1}{2} \int \frac{t}{1+t^2} \cdot 2t \, dt$$

$$\Rightarrow I = x \tan^{-1} \sqrt{x} - \int \frac{t^2}{1+t^2} \, dt$$

$$\Rightarrow I = x \tan^{-1} \sqrt{x} - \int \frac{1+t^2-1}{1+t^2} dt$$

$$\Rightarrow I = x \tan^{-1} \sqrt{x} - \int \left(1 - \frac{1}{1+t^2} \right) dt$$

$$\Rightarrow I = x \tan^{-1} \sqrt{x} - \int 1 dt + \int \left(\frac{1}{1+t^2} \right) dt$$

$$\Rightarrow I = x \tan^{-1} \sqrt{x} - t + \tan^{-1} t + C$$

$$\Rightarrow I = x \tan^{-1} \sqrt{x} - \sqrt{x} + \tan^{-1} \sqrt{x} + C$$

$$\Rightarrow I = (x+1) \tan^{-1} \sqrt{x} - \sqrt{x} + C$$

Hence, option (A) is correct answer.

51. $\int e^x \left(\frac{1-x}{1+x^2} \right)^2 dx$ is equal to

(A) $\frac{e^x}{1+x^2} + C$

(B) $\frac{-e^x}{1+x^2} + C$

(C) $\frac{e^x}{(1+x^2)^2} + C$

(D) $\frac{-e^x}{(1+x^2)^2} + C$

Ans: Let $I = \int e^x \left(\frac{1-x}{1+x^2} \right)^2 dx$

$$\Rightarrow I = \int e^x \left[\frac{1+x^2-2x}{(1+x^2)^2} \right] dx$$

$$\Rightarrow I = \int e^x \left[\frac{1+x^2}{(1+x^2)^2} - \frac{2x}{(1+x^2)^2} \right] dx$$

$$\Rightarrow I = \int e^x \left[\frac{1}{(1+x^2)} - \frac{2x}{(1+x^2)^2} \right] dx$$

$$\Rightarrow I = \frac{e^x}{(1+x^2)} + C$$

Hence, option (A) is correct answer.

52. $\int \frac{x^9}{(4x^2+1)^6} dx$ is equal to

(A) $\frac{1}{5x} \left(4 + \frac{1}{x^2} \right)^{-5} + C$

(B) $\frac{1}{5} \left(4 + \frac{1}{x^2} \right)^{-5} + C$

(C) $\frac{1}{10x} (1+4)^{-5} + C$

(D) $\frac{1}{10} \left(4 + \frac{1}{x^2} \right)^{-5} + C$

Ans: Let $I = \int \frac{x^9}{(4x^2+1)^6} dx$

$$I = \int \frac{x^9}{\left[x^2 \left(4 + \frac{1}{x^2} \right) \right]^6} dx$$

$$\Rightarrow$$

$$I = \int \frac{x^9}{x^{12} \left(4 + \frac{1}{x^2} \right)^6} dx$$

$$\Rightarrow$$

$$I = \int \frac{dx}{x^3 \left(4 + \frac{1}{x^2} \right)^6}$$

$$\Rightarrow$$

Now, put $4 + \frac{1}{x^2} = t$

$$\Rightarrow -\frac{2}{x^3} dx = dt$$

$$\Rightarrow \frac{dx}{x^3} = -\frac{dt}{2}$$

Therefore,

$$\Rightarrow I = -\frac{1}{2} \int \frac{dt}{(t)^6}$$

$$\Rightarrow I = -\frac{1}{2} \int t^{-6} dt$$

$$\Rightarrow I = -\frac{1}{2} \times \frac{t^{-5}}{-5} + C$$

$$\Rightarrow I = \frac{1}{10} \left(4 + \frac{1}{x^2} \right)^{-5} + C$$

Hence, option (D) is correct answer.

53. $\int \frac{dx}{(x+2)(x^2+1)} = a \log|1+x^2| + b \tan^{-1}x + \frac{1}{5} \log|x+2| + C$, then

(A). $a = -\frac{1}{10}, b = -\frac{2}{5}$

(B) $a = \frac{1}{10}, b = -\frac{2}{5}$

(C) $a = -\frac{1}{10}, b = \frac{2}{5}$

(D) $a = \frac{1}{10}, b = \frac{2}{5}$

Ans: Let $I = \int \frac{dx}{(x+2)(x^2+1)}$

Now, $\frac{1}{(x+2)(x^2+1)} = \frac{A}{(x+2)} + \frac{Bx+C}{(x^2+1)}$

$$\Rightarrow \frac{1}{(x+2)(x^2+1)} = \frac{A(x^2+1) + (Bx+C)(x+2)}{(x^2+1)}$$

$$\Rightarrow 1 = A(x^2+1) + (Bx+C)(x+2)$$

$$\Rightarrow 1 = Ax^2 + A + Bx^2 + 2Bx + Cx + 2C$$

$$\Rightarrow 1 = x^2(A+B) + x(2B+C) + (A+2C)$$

On comparing both sides, we get

$$\Rightarrow A+B=0, 2B+C=0 \text{ and } A+2C=1$$

From above equations, we get

$$\Rightarrow A = \frac{1}{5}, B = -\frac{1}{5} \text{ and } C = \frac{2}{5}$$

Therefore, $\frac{1}{(x+2)(x^2+1)} = \frac{1}{5(x+2)} + \frac{-\frac{x}{5} + \frac{2}{5}}{(x^2+1)}$

$$\Rightarrow \int \frac{dx}{(x+2)(x^2+1)} = \int \frac{dx}{5(x+2)} + \int \frac{-\frac{x}{5} + \frac{2}{5}}{(x^2+1)} dx$$

$$\Rightarrow \int \frac{dx}{(x+2)(x^2+1)} = \int \frac{dx}{5(x+2)} - \frac{1}{5} \int \frac{x}{(x^2+1)} dx + \frac{2}{5} \int \frac{1}{(x^2+1)} dx$$

$$= \frac{1}{5} \log|x+2| - \frac{1}{10} \log|1+x^2| + \frac{2}{5} \tan^{-1}x + C$$

Here, $a = -\frac{1}{10}$ and $b = \frac{2}{5}$

Hence, option (C) is correct answer.

54. $\int \frac{x^3}{x+1} dx$ is equal to

(A) $x + \frac{x^2}{2} + \frac{x^3}{3} - \log|1-x| + C$

(B) $x + \frac{x^2}{2} - \frac{x^3}{3} - \log|1-x| + C$

(C) $x - \frac{x^2}{2} - \frac{x^3}{3} - \log|1+x| + C$

(D) $x - \frac{x^2}{2} + \frac{x^3}{3} - \log|1+x| + C$

Ans: Let $I = \int \frac{x^3}{x+1} dx$

$$\Rightarrow I = \int \frac{x^3 + 1 - 1}{x + 1} dx$$

$$\Rightarrow I = \int \frac{x^3 + 1}{x + 1} dx - \int \frac{1}{x + 1} dx$$

$$\Rightarrow I = \int (x^2 - x + 1) dx - \int \frac{1}{x + 1} dx$$

$$\Rightarrow I = \frac{x^3}{3} - \frac{x^2}{2} + x - \log|x + 1| + C$$

Hence, option (D) is correct answer.

55. $\int \frac{x + \sin x}{1 + \cos x} dx$ is equal to

(A) $\log|1 + \cos x| + C$

(B) $\log|x + \sin x| + C$

(C) $x - \tan \frac{x}{2} + C$

(D) $x \tan \frac{x}{2} + C$

Ans: Let $I = \int \frac{x + \sin x}{1 + \cos x} dx$

$$\Rightarrow I = \int \frac{x}{1 + \cos x} dx + \int \frac{\sin x}{1 + \cos x} dx$$

$$\Rightarrow I = I_1 + I_2 \dots\dots\dots(i), \text{ where, } I_1 = \int \frac{x}{1 + \cos x} dx \quad \& \quad I_2 = \int \frac{\sin x}{1 + \cos x} dx$$

Now, $I_1 = \int \frac{x}{1 + \cos x} dx$

$$I_1 = \int \frac{x}{2 \cos^2 \frac{x}{2}} dx$$

$\Rightarrow$

$$\Rightarrow I_1 = \frac{1}{2} \int x \sec^2 \frac{x}{2} dx$$

$$\Rightarrow I_1 = \frac{1}{2} \left[x \int \sec^2 \frac{x}{2} dx - \int \left(\frac{d}{dx} x \cdot \int \sec^2 \frac{x}{2} dx \right) dx \right]$$

$$\Rightarrow I_1 = \frac{1}{2} \left[2x \tan \frac{x}{2} - 2 \int \tan \frac{x}{2} dx \right] + C$$

$$\Rightarrow I_1 = x \tan \frac{x}{2} - \int \tan \frac{x}{2} dx + C \quad \dots\dots\dots (ii)$$

And also we have,

$$\Rightarrow I_2 = \int \frac{\sin x}{1 + \cos x} dx$$

$$\Rightarrow I_2 = \int \frac{2 \sin \frac{x}{2} \cdot \cos \frac{x}{2}}{2 \cos^2 \frac{x}{2}} dx$$

$\Rightarrow$

$$\Rightarrow I_2 = \int \frac{\sin \frac{x}{2}}{\cos \frac{x}{2}} dx$$

$\Rightarrow$

$$\Rightarrow I_2 = \int \tan \frac{x}{2} dx \quad \dots\dots\dots (iii)$$

Now, put the value of I_1 and I_2 in eq (i),

$$\Rightarrow I = x \tan \frac{x}{2} - \int \tan \frac{x}{2} dx + C + \int \tan \frac{x}{2} dx$$

$$\Rightarrow I = x \tan \frac{x}{2} + C$$

Hence, option (D) is correct answer.

56. $\int \frac{x^3 dx}{\sqrt{1+x^2}} = a(1+x^2)^{\frac{3}{2}} + b\sqrt{1+x^2} + C,$ then

(A) $a = \frac{1}{3}, b = 1$

(B) $a = \frac{-1}{3}, b = 1$

(C) $a = \frac{-1}{3}, b = -1$

(D) $a = \frac{1}{3}, b = -1$

Ans: Let $I = \int \frac{x^3 dx}{\sqrt{1+x^2}}$

$\Rightarrow I = \int \frac{x^2 \cdot x dx}{\sqrt{1+x^2}}$

Put $1+x^2 = t^2$

$\Rightarrow 2x dx = 2t dt$

$\Rightarrow x dx = t dt$

Therefore,

$\Rightarrow I = \int \frac{(t^2 - 1)t dt}{t}$

$\Rightarrow I = \int (t^2 - 1) dt$

$\Rightarrow I = \frac{t^3}{3} - t + C$

$$\Rightarrow I = \frac{(\sqrt{1+x^2})^3}{3} - \sqrt{1+x^2} + C$$

$$\Rightarrow I = \frac{(1+x^2)^{\frac{3}{2}}}{3} - \sqrt{1+x^2} + C$$

Here, $a = \frac{1}{3}, b = -1$

Hence, option (D) is correct answer.

57 $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{dx}{1+\cos 2x}$ is equal to

(A) 1

(B) 2

(C) 3

(D) 4

Ans: Let $I = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{dx}{1+\cos 2x}$

$$\Rightarrow I = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{dx}{2\cos^2 x}$$

$$\Rightarrow I = \frac{1}{2} \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sec^2 x dx$$

$$\Rightarrow I = \frac{1}{2} [\tan x]_{-\frac{\pi}{4}}^{\frac{\pi}{4}}$$

$$\Rightarrow I = \frac{1}{2} \left[\tan \frac{\pi}{4} - \tan \left(-\frac{\pi}{4} \right) \right]$$

$$\Rightarrow I = \frac{1}{2} [1 + 1]$$

$$\Rightarrow I = 1$$

Hence, option (A) is correct answer.

58. $\int_0^{\frac{\pi}{2}} \sqrt{1 - \sin 2x} \, dx$ is equal to

(A) $2\sqrt{2}$

(B) $2(\sqrt{2} + 1)$

(C) 2

(D) $2(\sqrt{2} - 1)$

Ans: Let $I = \int_0^{\frac{\pi}{2}} \sqrt{1 - \sin 2x} \, dx$

$$\Rightarrow I = \int_0^{\frac{\pi}{2}} \sqrt{\sin^2 x + \cos^2 x - 2 \sin x \cos x} \, dx$$

$$\Rightarrow I = \int_0^{\frac{\pi}{2}} \sqrt{(\sin x - \cos x)^2} \, dx$$

$$\Rightarrow I = \int_0^{\frac{\pi}{2}} |\sin x - \cos x| \, dx$$

Since, $(\sin x - \cos x) \leq 0$, $x \in \left[0, \frac{\pi}{4} \right]$ and $(\sin x - \cos x) \geq 0$, $x \in \left[\frac{\pi}{4}, \frac{\pi}{2} \right]$

Therefore,

$$\Rightarrow I = -\int_0^{\frac{\pi}{4}} (\sin x - \cos x) dx + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (\sin x - \cos x) dx$$

$$\Rightarrow I = -\left[-\cos x - \sin x\right]_0^{\frac{\pi}{4}} + \left[-\cos x - \sin x\right]_{\frac{\pi}{4}}^{\frac{\pi}{2}}$$

$$\Rightarrow I = \left[\cos x + \sin x\right]_0^{\frac{\pi}{4}} - \left[\cos x + \sin x\right]_{\frac{\pi}{4}}^{\frac{\pi}{2}}$$

$$\Rightarrow I = \left[\cos \frac{\pi}{4} + \sin \frac{\pi}{4} - 1 - 0\right] - \left[\cos \frac{\pi}{2} + \sin \frac{\pi}{2} - \cos \frac{\pi}{4} - \sin \frac{\pi}{4}\right]$$

$$\Rightarrow I = \left[\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} - 1\right] - \left[1 - \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}\right]$$

$$\Rightarrow I = [\sqrt{2} - 1] - [1 - \sqrt{2}]$$

$$\Rightarrow I = \sqrt{2} - 1 - 1 + \sqrt{2}$$

$$\Rightarrow I = 2\sqrt{2} - 2$$

$$\Rightarrow I = 2(\sqrt{2} - 1)$$

Hence, option (D) is correct answer.

59. $\int_0^{\frac{\pi}{2}} \cos x e^{\sin x} dx$ is equal to

Ans: Let $I = \int_0^{\frac{\pi}{2}} \cos x e^{\sin x} dx$

Put $\sin x = t$

$$\Rightarrow \cos x dx = dt$$

Now, at $x = 0 \Rightarrow t = 0$ and at $x = \frac{\pi}{2} \Rightarrow t = 1$

Therefore,

$$\Rightarrow I = \int_0^1 e^t dx$$

$$\Rightarrow I = [e^t]_0^1$$

$$\Rightarrow I = e^1 - e^0$$

$$\Rightarrow I = e - 1$$

Hence, $\int_0^{\frac{\pi}{2}} \cos x e^{\sin x} dx = e - 1.$

60. $\int \frac{x+3}{(x+4)^2} e^x dx = \dots\dots\dots$

Ans: Let $I = \int \frac{x+3}{(x+4)^2} e^x dx$

$$\Rightarrow I = \int e^x \left[\frac{x+4-1}{(x+4)^2} \right] dx$$

$$\Rightarrow I = \int e^x \left[\frac{x+4}{(x+4)^2} - \frac{1}{(x+4)^2} \right] dx$$

$$\Rightarrow I = \int e^x \left[\frac{1}{x+4} - \frac{1}{(x+4)^2} \right] dx$$

$$\Rightarrow I = \frac{e^x}{x+4} + C$$

Hence, $\int \frac{x+3}{(x+4)^2} e^x dx = \frac{e^x}{x+4} + C.$

Fill in the blanks in each of the following Exercise 60 to 63.

61. $\int_0^a \frac{1}{1+4x^2} dx = \frac{\pi}{8}$, then $a = \dots\dots\dots$

Ans: here, we have $\int_0^a \frac{1}{1+4x^2} dx = \frac{\pi}{8}$

$$\Rightarrow \frac{1}{4} \int_0^a \frac{1}{\frac{1}{4} + x^2} dx = \frac{\pi}{8}$$

$$\Rightarrow \int_0^a \frac{1}{\left(\frac{1}{2}\right)^2 + x^2} dx = \frac{\pi}{2}$$

$$\Rightarrow \left[\frac{1}{\frac{1}{2}} \tan^{-1} \left(\frac{x}{\frac{1}{2}} \right) \right]_0^a = \frac{\pi}{2}$$

$$\Rightarrow \left[2 \tan^{-1}(2x) \right]_0^a = \frac{\pi}{2}$$

$$\Rightarrow \tan^{-1}(2a) - \tan^{-1}(0) = \frac{\pi}{4}$$

$$\Rightarrow \tan^{-1}(2a) = \frac{\pi}{4}$$

$$\Rightarrow 2a = \tan \frac{\pi}{4}$$

$$\Rightarrow 2a = 1$$

$$\Rightarrow a = \frac{1}{2}$$

Hence, value of a is $\frac{1}{2}$.

Therefore, $\int_0^a \frac{1}{1+4x^2} dx = \frac{\pi}{8}$, then $a = \frac{1}{2}$.

62. $\int \frac{\sin x}{3+4 \cos^2 x} dx =$

Ans: Let $I = \int \frac{\sin x}{3+4 \cos^2 x} dx$

Put $\cos x = t$

$$\Rightarrow -\sin x dx = dt$$

$$\Rightarrow \sin x dx = -dt$$

Therefore,

$$\Rightarrow I = -\int \frac{dt}{3+4t^2}$$

$$\Rightarrow I = -\frac{1}{4} \int \frac{dt}{\frac{3}{4}+t^2}$$

$$\Rightarrow I = -\frac{1}{4} \int \frac{dt}{\left(\frac{\sqrt{3}}{2}\right)^2+t^2}$$

$$\Rightarrow I = -\frac{1}{4} \cdot \frac{1}{\frac{\sqrt{3}}{2}} \tan^{-1} \left(\frac{t}{\frac{\sqrt{3}}{2}} \right) + C$$

$$\Rightarrow I = -\frac{1}{2\sqrt{3}} \tan^{-1} \left(\frac{2t}{\sqrt{3}} \right) + C$$

$$\Rightarrow I = -\frac{1}{2\sqrt{3}} \tan^{-1} \left(\frac{2\cos x}{\sqrt{3}} \right) + C$$

Hence, $\int \frac{\sin x}{3+4\cos^2 x} dx = -\frac{1}{2\sqrt{3}} \tan^{-1} \left(\frac{2\cos x}{\sqrt{3}} \right) + C.$

63. The value of $\int_{-\pi}^{\pi} \sin^3 x \cdot \cos^2 x dx$ is

Ans: Let $I = \int_{-\pi}^{\pi} \sin^3 x \cdot \cos^2 x dx$ (i)

$$\Rightarrow I = \int_{-\pi}^{\pi} \sin^3 (\pi - \pi - x) \cdot \cos^2 (\pi - \pi - x) dx$$

$$\Rightarrow I = \int_{-\pi}^{\pi} \sin^3 (-x) \cdot \cos^2 (-x) dx$$

$$\Rightarrow I = -\int_{-\pi}^{\pi} \sin^3 x \cdot \cos^2 x dx \text{(ii)}$$

Now, adding eq (i) and (ii), we get

$$\Rightarrow 2I = 0$$

$$\Rightarrow I = 0$$

Hence, the value of $\int_{-\pi}^{\pi} \sin^3 x \cdot \cos^2 x dx$ is 0.